

AN ALMA ARCHIVAL SEARCH FOR SPECTRAL TECHNOSIGNATURES TOWARD STARS WITHIN 40 PC

GLENN J. WHITE^{1,2} AND ROBIN DEY³

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ABSTRACT

We have searched 431 archival ALMA spectral windows spanning 89.6–495.1 GHz (Bands 3–8) toward 88 stars in 81 systems within 40 pc for unresolved spectral carriers, comparing each stellar position with 512 control positions in an annulus centred on the star. **No credible technosignature is found.** The sample is every star that usable public archival observations cover, so it is ALMA-archive-complete within 40 pc rather than interest-selected, but it is not a volume-complete stellar sample: it holds ~ 1 per cent of the stars in that volume, over-representing F and G types ($\times 3.7$, $\times 3.1$) and under-representing M dwarfs ($\times 0.6$). Everything above ~ 115 GHz is essentially unsearched elsewhere, and these are the first millimetre limits for Barnard’s Star and Wolf 359. Four windows raise a statistical screening flag rather than a candidate detection: two toward β Pic and one toward HD 48370 are consistent with known circumstellar or foreground CO, and one, toward CP–72 2713, is unclassified. The screening test resolves no probability below $1/513$, far above the survey-wide Bonferroni scale 1.2×10^{-4} , so no flag is by itself significant. A second-stage local null resolves past that floor: for CP–72 2713 it gives $p = 1.6 \times 10^{-3}$, confirmed at $p = 1.2 \times 10^{-3}$ by an independent extreme-value fit to the measured controls, about 13 times above the Bonferroni scale. Trigger thresholds, the powers at which a signal first exceeds the screening test, span 2.3×10^{13} – 6.8×10^{16} W EIRP at 15.3–1953 kHz native resolution over the 118 drift-resolved Class A windows, and 1.6×10^{13} – 2.8×10^{17} W at 15.625–31.25 MHz over the 313 unresolved-spectral-excess Class B windows; the frequency-matched benchmark is a 12 m aperture radiating 1 MW at 230 GHz, EIRP $\approx 8 \times 10^{14}$ W. 313 of the 431 windows (73 per cent) carry no drift discrimination. Two limits qualify every threshold quoted here: ALMA’s default Hanning response leaves 0.38–0.50 of a sub-channel carrier’s power in the peak channel, so the thresholds are optimistic by $\times 2.00$ to $\times 2.7$; and completeness for the drift-resolved class is measured on a single configuration and transferred by assumption, with unbounded transfer error. The detection statistic and the line mask were defined using these data, so the rank probabilities and false-alarm figures are descriptive of this dataset rather than an independent validation of the pipeline’s false-positive rate.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

A technological civilisation, signalling deliberately or going about its affairs, may leave a detectable imprint on the electromagnetic spectrum: a *technosignature* (Sheikh 2020; Vidal et al. 2026). Six decades of radio searches cover many thousands of stars yet, as Wright et al. (2018) quantified, sample only 6×10^{-18} of the multi-dimensional haystack of transmitter frequency, bandwidth, power and duty cycle. Three gaps persist: the millimetre and submillimetre regime above ~ 50 GHz is essentially unsearched; most surveys target interest-selected classes of star, not the local stellar population; and few reprocess *existing* interferometric archives.

The symmetric detection statistic and the ± 50 km s^{−1} molecular-line mask were both defined, and the statistic once revised, using these data (§5.4). The rank probabilities and false-alarm figures below are therefore descriptive of this dataset, not an independent validation of the pipeline’s false-positive rate; this is restated where it bears on a number.

This paper addresses all three, asking how strongly existing ALMA observations constrain unresolved artificial spectral emission from nearby stellar systems. What follows characterises the stars ALMA has observed, not the 40 pc population: the searched subsample over-represents F and G stars and under-represents M dwarfs (§3), and every limit below inherits that bias. The sample is ALMA-archive-complete within 40 pc: every star in that volume with public ALMA archival coverage, whatever its spectral type, disc-hosting status or known planets. We mine it, correcting for proper motion and Doppler drift, for unresolved spectral carriers (channel-confined excess at native ALMA resolution) and anomalous millimetre continuum. Throughout, “narrowband” describes the assumed intrinsically narrow transmitter and never the resolving power of the data (§4). The processing sets the search class, measured in §4.4: a continuous carrier is the best case, short-duty fast-drifting emitters the worst. Coarse-channel windows retain sensitivity to persistent unresolved excess power only (§4). Three contributions carry the paper: systematic exploitation of archival ALMA stellar fields, 107 target/band datasets mined from public data with no new observing time; spatial-control null

¹ School of Physical Sciences, The Open University, Walton Hall, Milton Keynes, MK7 6AA, UK; glenn.white@open.ac.uk

² RAL Space, STFC Rutherford Appleton Laboratory, Harwell Campus, Didcot, OX11 0QX, UK

³ VBRL Holdings Inc.

calibration for interferometric SETI, the same statistic evaluated at the star and at 512 control positions so the false-alarm rate is measured, not assumed; and a quantitative accounting of the frequency, EIRP and drift domain actually searched, which turns a non-detection into a statement with edges. Of the ancillary analyses (closure-phase vetting, a continuum anomaly screen, a chirp and periodicity re-analysis, a cross-target frequency-occupancy check), only the continuum screen enters a disposition; the rest are capability demonstrations (§4.3). One worked case runs through the paper: a crossing towards β Pictoris, the largest star-versus-control contrast here, attributed to the disc’s known circumstellar CO across two transitions and three epochs (§5.3). Per-target results are in Appendix E; dispositions, false-alarm accounting and injection completeness consolidate in Tables 5 and 6 with Appendices I and B.

2. BACKGROUND

Six decades of searching, from Cocconi & Morrison (1959) and Drake (1960) through the programmes reviewed by Tarter (2001) and Project Phoenix (Backus & Project Phoenix Team 2002), divide into wide-field commensal and targeted surveys (Zhang et al. 2020; Czech et al. 2021; Morrison et al. 2023; Li et al. 2026; Johnson et al. 2023; Valente et al. 2025; Tremblay et al. 2024), with machine-learning interference rejection now central (Ma et al. 2023; Choza et al. 2024; Jacobson-Bell et al. 2025; Poznanski et al. 2025; Garrett et al. 2026), a capability our classical threshold search lacks. Essentially all of it lies below ~ 10 GHz. Above it the mm/submm regime has been searched once, also with ALMA (Mason et al. 2024), despite physical reasons to go higher: interstellar scattering and dispersion fall steeply with frequency, a given aperture delivers a tighter beam per watt of EIRP, and directional communication, radar and beamed power transmission are already engineered coherent mm-wave emission, so plausibility needs no assumption of Arecibo-scale transmitters at hundreds of gigahertz. The countervailing costs, atmospheric transmission, receiver noise, photons per watt per hertz and pointing and phase stability, price the search, not forbid it.

Mason et al. (2024) searched archival Band 3 windows toward 28 “stellar bycatch” stars, observed for other science and usable for free, placing $\text{EIRP}_{\min} > 7 \times 10^{17}$ W for the nearest and identifying the challenges of high-frequency work: drift rates scale with frequency, smearing erodes sensitivity without a drift search, and the molecular line forest confuses. Our pilot (White 2026, citation provisional) extended that methodology to TRAPPIST-1 across Bands 3, 6 and 7 ($\text{EIRP}_{\min} \sim 5\text{--}10 \times 10^{13}$ W, no detection) and built the ranked 100-star list this survey grew from. Its pipeline is that design applied to one target. Nothing in this paper depends on that pilot: every claim attributed to it below is restated from the products released with this work. New here are the archive-complete sample selection (§3), the 512-position control annulus and its local false-alarm estimation, closure-phase vetting, the spectral-type-normalised continuum screen, the chirp and periodicity extension and the cross-target frequency-occupancy check. Each is new, to our knowledge, *in an ALMA archival technosignature survey*; we claim no wider priority over their centimetre-wave counterparts. No execution block is analysed in both papers: the pilot’s 100-star search products are superseded here, not re-used, and every window here was re-reduced and re-searched from the raw archive data with the pipeline of §4. Where a star is treated in both, the numbers to cite are this paper’s frozen products, and any discrepancy resolves in their favour.

Figure 1 places the nominal EIRP thresholds among published searches in *total power*. As the in-panel note warns, vertical position is not technosignature merit: this survey’s competitive EIRP toward the nearest systems is predominantly the d^2 factor (d spans a factor $\sim 10^3$ in d^2) rather than sensitivity to Hz-wide carriers, to which a 15.625-MHz channel and a 2.8-Hz SETI channel respond very differently (§4).

A non-detection must become a quantitative exclusion statement: Wright et al. (2018) formalised the “Cosmic Haystack” search volume, Margot et al. (2023) its conversion into an upper bound on the transmitter-hosting fraction, and Sheikh et al. (2025a) asked where Earth’s technosignatures would be detectable. That bound is valid only if the sample represents the underlying population, which interest-selected samples do not. Most surveys select by proximity, spectral type, planets or Earth Transit Zone membership; the ALMA-specific form, disc selection, skews the sample young and dynamically active. We instead take every star within 40 pc in the usable field of at least one public ALMA observation (§3): archive-complete for the observed subset of the local population in a fixed volume, not for that population itself. This removes technosignature-specific but not archival astrophysical selection; the 168 sample entries are ~ 1 per cent of the catalogued population within 40 pc (the 88 actually searched here are ~ 0.5 per cent), and the residual coverage selection is characterised as a completeness function (§3, Table 5).

3. SAMPLE SELECTION

The sample is ALMA-archive-complete within 40 pc, which is not a volume-complete stellar sample: it is every star within 40 pc of the Sun (Gaia DR3 parallaxes, distances direct inversions $d = 1/\pi$; every entry $\pi \geq 25$ mas with $\sigma_\pi/\pi \leq 0.8$ per cent) whose proper-motion-corrected position at the observation epoch falls in the field of view of at least one public ALMA execution block, not necessarily at the pointing centre: mosaics and wide pointings serendipitously cover more than one catalogued star. Disc, planet and spectral type are descriptive metadata here, not selection criteria. Throughout, a *star* is one catalogue entry and a *system* the set of entries bound to it; a *target/band* dataset is one star in one ALMA band; an *execution block* (EB) one archival observation; and a *window* is one spectral window of one execution block. This gives 168 unique target stars, processed nearest-first. Because trigger thresholds scale with d^2 at fixed noise, that order is sensitivity-ranked, not an unbiased draw, and the prevalence statements of §6 are conditional on it.

Composition follows from the archival selection, tabulated against the reference census in Table 10. It rests on `teff_gspphot` as a spectral-class proxy, available for 52 of 88 sample stars and 49 per cent of the census and missing preferentially for the coolest objects, so the M fraction is a lower limit and the earlier-type ratios upper limits: of

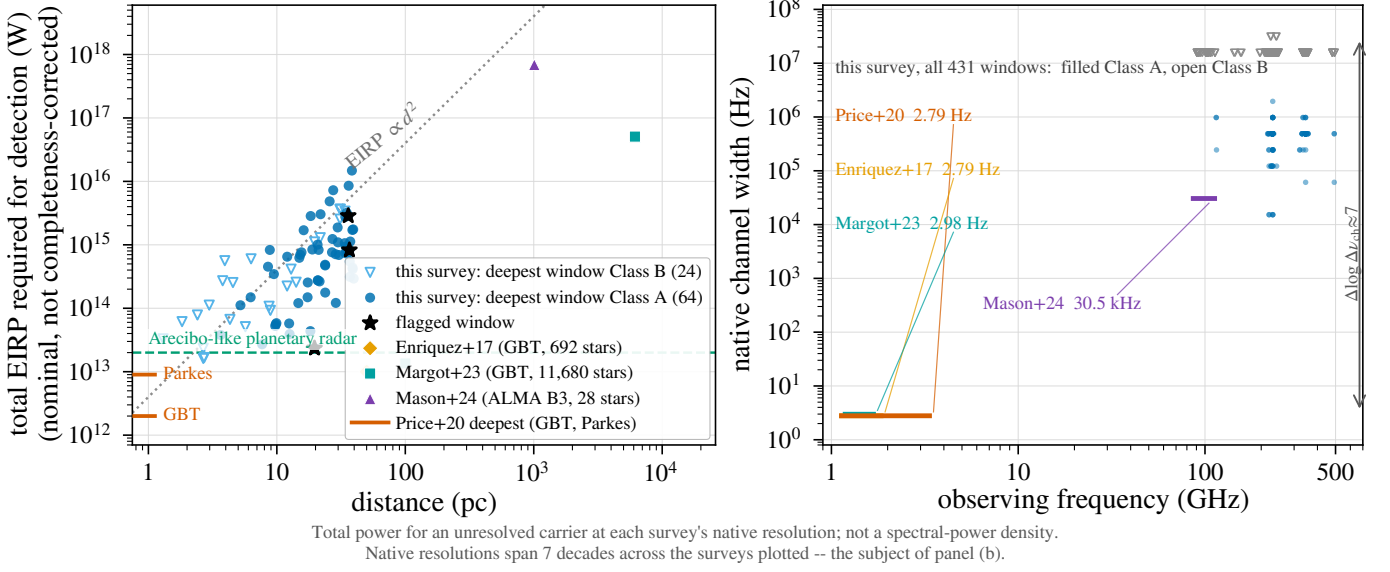


FIG. 1.— Literature context of the 5σ trigger thresholds, in total power (§2). (a) EIRP against distance. Blue: this survey, deepest window per star (88 stars, 1.30–39.63 pc; filled: Class A, drift-resolved; open: Class B, unresolved spectral excess); orange: Enriquez et al. (2017); aqua: Margot et al. (2023); violet: Mason et al. (2024); left-edge ticks: deepest per-target limits of Price et al. (2020). Grey dotted: $EIRP \propto d^2$ at constant received flux. Green dashed: Arecibo-class power (§4.2). (b) Native channel width against observing frequency for the same windows and programmes: native resolutions differ by $\sim 10^7$. Thresholds are nominal (boxed rule, §4); conventions differ among these surveys.

the searched stars carrying a Gaia temperature, 41 per cent are M dwarfs against 69 per cent of the reference census, while F and G stars are over-represented (Table 10). The archive prefers earlier types and nearer systems, so the non-detection bears on the ALMA-observed, F/G-enriched mix of Table 10, which under-represents the M dwarfs dominating the local volume: a mismatch of astrobiological priority, not of statistical validity. Eighteen of the 88 processed stars are confirmed exoplanet hosts (§6); the debris-disc hosts (β Pic, AU Mic, ϵ Eri, τ Cet) contribute dusty continua, kept from masquerading as anomalies by the continuum lane’s spectral-type-normalised screen (§4). Window counts are also concentrated: β Pic (15 windows), AU Mic (12) and TRAPPIST-1 (12) alone carry 39 of the 431 searched windows (9.0 per cent) from 3 systems, so any window-weighted statement leans on a few heavily observed young, active or disc-hosting systems.

The 168 stars are complete with respect to, and conditional only on, the 40-parsec reference catalogue intersected with the public-archive snapshot of 2026 September 8. Against that Gaia DR3 solar-neighbourhood catalogue (17 566 stars within 40 pc) they are ~ 1 per cent. The two parallax cuts are not interchangeable: the sample requires $\sigma_\varpi/\varpi \leq 0.8$ per cent, the reference census ≤ 10 per cent (Table 10), so the ratio compares two differently selected populations; the census is also incomplete at both ends, and the fraction says only that coverage is small. Table 10 leaves the selection function inspectable; age and disc-hosting flags are not uniformly available and are documented qualitatively.

Stellar counts throughout are catalogue-entry counts. The 88 searched entries comprise 81 independent systems, differing by seven designation-linked component pairs (SIMBAD designations; per-entry rows, distances and unresolved-pair notes in the machine-readable catalogue). The 107 star/band datasets, 431 windows and 102 execution blocks therefore derive from 81 systems, the paired rows being bookkeeping, not independent trials.

“Usable public ALMA coverage” is a five-criterion decision tree, frozen absent an erratum: (i) at least one public execution block (EB) whose QA2-calibrated visibilities cover the star’s epoch-propagated position in the primary beam, with no minimum integration, sensitivity, antenna count, mosaic-tile count or array configuration beyond QA2; (ii) that position within one primary-beam FWHM of the field centre (largest accepted offset ~ 0.7 FWHMs, Sirius B, retained with bounded 3.6–16 per cent primary-beam corrections whose residual systematic a holography-beam recomputation would settle, §4.2); (iii) each spectral window admitted individually, on QA2 vetting alone; (iv) no programme-category restriction, science, calibration and observatory projects entering on identical terms; (v) a fixed archive snapshot date (2026 September 8). No star or window was excluded by an unlisted criterion, none for short integration (the shortest retained are 1.0 and 1.5 min).

ALMA project metadata quantify pointed versus serendipitous coverage: almost every spectral-window-covered star is the pointed science target in at least one execution block, the exceptions a few blocks on another programme’s field or calibrator-only, leaving Wolf 219 the sole serendipitous star. An archive-complete sample’s residual selection function is the union of ALMA target lists, differing from an interest-selected sample in degree, but the degree is *quantified*: a bycatch fraction of 1 star in 44, 1 EB in 57.

4. DATA AND METHODOLOGY

The design is one shared archive-ingestion stage feeding two independent search lanes over the 40 pc archive-complete sample; Table 3 gives the data path of the primary lane in order. For each target we mine, within a bounded budget, all calibrated (or locally recalibrated) archival execution blocks covering the star’s Gaia-propagated position, and

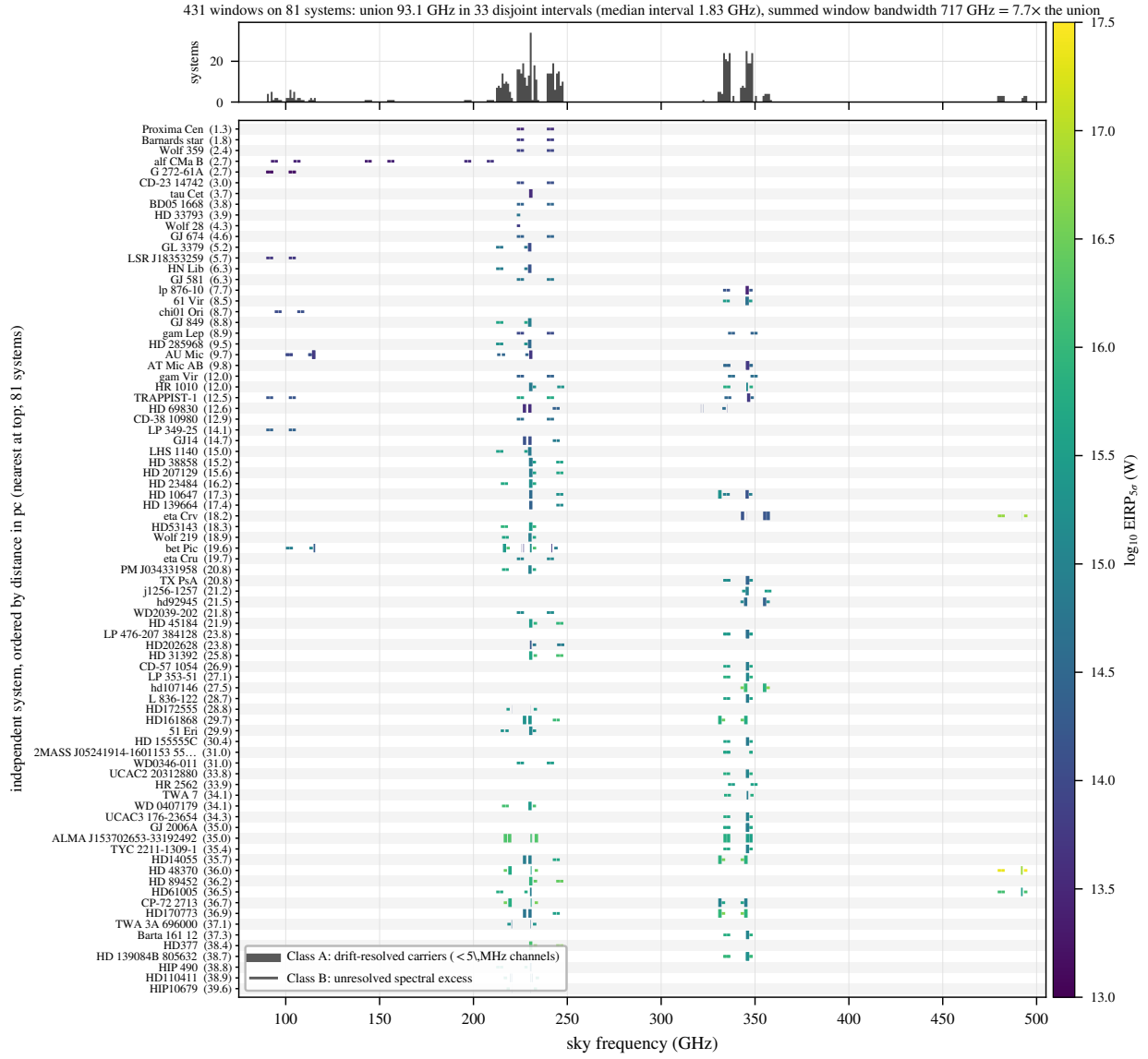


FIG. 2.— What this survey sampled in frequency. Each searched window is a horizontal segment on its host system’s row, rows ordered by distance, coloured by nominal $\text{EIRP}_{5\sigma}$, with line weight distinguishing the drift-resolved (Class A) from the unresolved-spectral-excess (Class B) channelisation; the upper panel counts systems per gigahertz of sky frequency. The coverage is narrow, clustered and repeatedly revisited: 33 islands concentrated near 230 and 345 GHz, with summed window bandwidth $7.6\times$ the union, which is what the in-band transmitter fractions of §6 condition on (Scope of the constraints, §4). Thresholds are nominal in the sense of §4.

search their visibilities in (i) the *primary* lane, a Doppler-drift-corrected spectral-carrier search following Mason et al. (2024) and Enriquez et al. (2017), converting each non-detection into an equivalent isotropic radiated power limit $\text{EIRP}_{5\sigma} = 4\pi d^2 S_{\min} \Delta\nu$, with $S_{\min}$ five times the searched window’s per-channel rms, primary-beam corrected, $\Delta\nu$ its native channel width and d the target distance (worked numerically in Appendix A). The quoted thresholds are nominal. Their systematic budget is the ALMA absolute flux scale (5–10 per cent, band- and epoch-dependent), the primary-beam model at the stellar offset, the instrumental spectral response (ALMA’s default online Hanning smoothing leaves 0.50 of a channel-centred sub-channel carrier’s power in the peak channel and 0.38 for one on a channel boundary, so every threshold quoted here is optimistic by $\times 2.00$ to $\times 2.7$, median $\times 2.29$; Appendix A) and visibility calibration. Each multiplies $S_{\min}$ and hence $\text{EIRP}_{5\sigma}$ directly; *Gaia* distance errors are negligible here. The same visibilities feed (ii) an *ancillary* line-excluded continuum lane (Appendix H). “ 5σ ” throughout is the local instrumental *trigger* threshold of Table 1, estimated empirically (§5.3). In the majority of windows an intrinsically Hz-wide carrier is unresolved within a 15.625 MHz channel, and the data could not demonstrate that a one-channel excess is intrinsically narrow. The search is reported as two formal survey classes, distinguished by the dimensionless drift parameter

$$\eta_{\text{drift}} = |\dot{\nu}_{\max}| T / \Delta\nu_{\text{ch}}, \quad (1)$$

the channel-width fraction the *maximum* searched drift $|\dot{\nu}_{\max}|$ traverses over an integration T at native channel width $\Delta\nu_{\text{ch}}$. A window is *drift-resolving* when $\eta_{\text{drift}} \gtrsim 1$, only then carrying a carrier across enough channels for its motion

to be measured, not assumed. The drift ceiling inside η_{drift} is a search-grid boundary, the widest trial drift the stack evaluates, not a physical prior on transmitter drift rates: carriers drifting faster are missed, and the resulting selection function is explicit in §6.1 (Table 5, Fig. 8).

Class A is the drift-resolved spectral-carrier search: the 118 windows of fine channelisation, native widths 15.259–1953.125 kHz (median 488.3 kHz; 62 per cent at the 488.3-kHz mode), spanning $\eta_{\text{drift}} = 0.34$ –422 (median 12.8). It is the only class with measured end-to-end recovery of a drifting carrier. *Class B* is the unresolved spectral-excess search: the 313 coarse windows, 73 per cent of the sample, native widths 15.625–31.25 MHz (309 windows at the 15.625-MHz mode), spanning $\eta_{\text{drift}} = 0.004$ –1.35 (median 0.36). It is sensitive to persistent near-stationary excess power, carries no drift discrimination, and is never described as a narrowband search anywhere in this paper.

η_{drift} is the physical criterion and the classes are labelled by the instrument. Channelisation is bimodal, with no window between 1953.125 kHz and 15.625 MHz, an empty gap of factor 8.0, so any boundary inside the gap assigns identical membership and figures label the classes by name. The two criteria do not coincide exactly, and the overlap is quantified: 12 long-integration Class B windows reach $\eta_{\text{drift}} \geq 1$, and only marginally (maximum 1.35), where a one-to-two channel traverse supports no discriminative drift fit after baseline removal; 6 short Class A windows fall below unity and are conservatively kept in Class A. Each window’s η_{drift} , η_{smear} , trial drift count and maximum searched acceleration $a_{\text{max}} = 3.6$ –4.0 m s^{−2} are columns of the released catalogue, so any reader may reclassify on the physical criterion alone. Results are given per class first and combined only where statistically appropriate (§5.3).

Both lanes exclude catalogued molecular transitions at tolerances suited to their statistics. The continuum lane masks the common species at native resolution before integrating; the carrier lane applies a ± 50 km s^{−1} stellar-frame velocity exclusion mask (§5.3) to any threshold crossing. The mask is *excluded search space*: the unmasked survey covers 93.1 GHz of unique sky frequency, the effective default quoted throughout is the ≈ 88.2 GHz surviving it, and no constraint is placed on carriers inside the excluded velocity regions. Masking reduces the frequency space a candidate can occupy, not the per-channel noise, so no threshold moves. All flux densities and EIRP thresholds are total intensity (Stokes I), formed from the two parallel-hand cross-correlations the archive delivers; no cross-hand ($V/Q/U$) extraction or circularity test was performed. The construction is polarisation-blind in amplitude, I being the summed parallel-hand power, so at fixed total transmitter power a strongly polarised artificial signal responds like an unpolarised one to within cross-polarisation leakage. That holds for the estimator, not the data: polarisation-dependent flagging in the archival QA2 calibration could have reduced sensitivity for some datasets, and the retained products do not record per-correlation flagging (the extraction sums over the correlation axis with QA2 weights), so we cannot state what fraction of the processed datasets retained both correlation hands unflagged through QA2. A dataset that lost one hand would be $\sqrt{2}$ less sensitive than its quoted threshold implies, and nothing here would catch it. Full Stokes analysis would add discrimination against weakly polarised backgrounds, not depth.

How to read every limit in this paper. *Every EIRP_{5 σ} value here is a nominal, local instrumental detection threshold, the trigger power P_{trig} : five times the per-channel rms of the searched window, at that window’s native channel width, for a transmitter at the star’s position matching the searched spectral morphology, with a measured recovery rising with the fraction of the track the carrier occupies its channel, highest for a continuous one (§4.3, Appendix B). It is not a calibrated completeness limit: completeness is the power P_x at which recovery reaches x per cent. $P_{50} = 1.10 P_{\text{trig}}$ and $P_{90} > 2 P_{\text{trig}}$ for the calibration configuration’s drifting class, both carrying an unbounded transfer systematic to any other configuration (bracketed as 0.5–2 $\times$, Appendix K), where they are not determined; P_{trig} is quoted throughout. It is a total-power threshold set by ALMA’s native channelisation, not a Hz-resolution sensitivity, optimistic by $\times 2.00$ to $\times 2.7$ under the instrument’s Hanning response (Appendix A). Every headline EIRP is quoted beside its native channel width, and the machine-readable catalogue carries the nominal threshold and both Hanning-corrected values per window. The same reading applies to every subsequent “5 σ ”.*

The three detectable morphologies, quantified with real windows in §6.1: a continuous frequency-stationary carrier (best case, near-ideal gain toward the nearest systems); a continuously drifting carrier in fine channels (42 per cent at threshold, 75–83 per cent at twice it); an intermittent duty-1/4 carrier (91 per cent fine, 100 per cent coarse channels).

4.1. The search statistic, and the words used for its output

The primary statistic is formed on extracted spectra, not images or visibilities. The extraction operates on the QA2-calibrated measurement set as delivered: per integration and per native channel, with the archive’s QA2 flagging carried through untouched and no spectral regridding or interpolation (Table 3, step 3); it returns the Stokes I intensity at the propagated sky position $\mathbf{x}$ (§4). Baseline removal is applied *after* extraction (step 4), and the primary-beam correction enters the quoted thresholds and fluxes (§4.2), not the extracted amplitudes the statistic ranks. The estimator’s visibility-domain weighting is fixed in the released code (Data Availability), which itemises the quantities consumed and emitted per window and is the reproducibility reference for steps 1–3. For a sky position $\mathbf{x}$, each per-integration spectrum is baseline-subtracted by a running median along the *frequency* axis. No temporal median is applied at any stage (§4.3). The residual dynamic spectrum is stacked along each trial linear drift rate $\dot{\nu}$, and the statistic is the

TABLE 1
NOMENCLATURE FOR SEARCH OUTPUT AND THE TWO SENSES IN WHICH A PROBABILITY IS QUOTED.

Term	Meaning
crossing	one channel $\times$ drift cell at the stellar position with $T \geq 5$: a threshold crossing (state 1)
stage-1 statistical flag	a window whose T_* exceeds all 512 control statistics (“star-exceeds-ring”): a screening device, not a candidate significance (state 2)
candidate	a stage-1 statistical flag surviving every veto; none here (state 3)
5σ trigger threshold (P_{trig})	local instrumental trigger level, <i>not</i> a completeness limit
completeness (P_x)	the EIRP at which recovery of the searched class reaches x per cent. P_{50} and P_{90} are undetermined outside the calibration configuration and carry an unbounded transfer systematic, bracketed 0.5–2 $\times$
$p_{\text{rank}, \text{min}}$	$1/(N_{\text{ctrl}} + 1) = 1/513$, the rank <i>resolution</i> of one window’s control ensemble: the smallest per-window probability the design can express
$P(\text{false flag})$	the survey-level false-positive probability after thresholding, correlation and non-Gaussianity. The two are not the same: the first is a property of one window’s ensemble, the second of the whole survey, and the survey-wide Bonferroni scale 1.2×10^{-4} lies two orders of magnitude below the first (§5.3)
crosswalk	“stage-1 statistical flag” (star-exceeds-ring) $\approx$ the “hit”/“event” of the wider literature (Enriquez et al. 2017; Margot et al. 2023) after their spatial-veto stage; the SP statistic $\approx$ their peak signal-to-noise ratio, with the false-alarm probability calibrated empirically by the control ring, not a Gaussian tail
Class A / Class B	drift-resolved spectral-carrier search (fine channels) / unresolved excess-power coarse search (§4)
f_{dwell}	intra-track dwell fraction: the fraction of one continuous observation a carrier occupies its channel
p_{epoch}	epoch activity probability: the chance a transmitter is on during a given analysed epoch (Appendix K)
ALMA-archive-complete	every star within 40 pc with public ALMA archival coverage (168 entries): complete within the archive, not a census of the stellar population
QA2	the ALMA pipeline’s second-stage quality-assurance products, used here without further observatory calibration
control ring	the 512 off-source test positions of one window, an annulus concentric with the star (§4.1)

largest single-channel amplitude in units of the window’s noise,

$$T(\mathbf{x}) = \max_{\nu, \nu} \frac{S(\mathbf{x}, \nu, \dot{\nu})}{\sigma(\mathbf{x})}, \quad (2)$$

where S is the drift-stacked flux density in one native channel and $\sigma(\mathbf{x})$ a single robust scale per position and window, estimated as $1.4826 \times$ the median absolute deviation of that position’s baseline-subtracted spectrum over all channels: local in position, global in frequency. The baseline median’s width W is likewise a single fixed channel count for all 431 windows, independent of native channel width, so the linewidth ceiling it imposes scales in frequency with each window’s channelisation (Appendix A). The crossing channel is *not* excluded from the scale, standard for a median-based estimator, whose 50-per-cent breakdown point makes a single-channel feature immaterial over a window’s hundreds of channels; the per-window local recomputation below excludes it explicitly. $T_* \equiv T(\mathbf{x}_*)$ at the proper-motion-propagated stellar position, and the same operation, drift grid, channelisation and noise estimator give $\{T_i\}$ at all 512 control positions.

The control geometry, in full. The controls are placed by `probe_positions()` in the released extraction code, which adds its offsets to the star’s own direction cosines, so the construction is concentric with the *star*, not the field phase centre. It is an annulus rather than a ring: 512 positions drawn uniform in area, $r = \sqrt{U(r_{\text{in}}^2, r_{\text{out}}^2)}$, and uniform in azimuth, from the fixed seed 20260825, so the layout is reproducible. Radii run 0.14–0.78 θ_{PB} with $\theta_{\text{PB}} = 1.22 \lambda/D$ evaluated at each window’s own median frequency, scaling the annulus window by window: across the survey θ_{PB} spans 12.7–69.5 arcsec, r_{in} 1.8–9.7 arcsec and r_{out} 9.9–54.2 arcsec (Table 2). A separate inner check-ring of 135 positions, reaching 0.06 θ_{PB} , serves the point-source test. Ring centre, θ_{PB} , r_{in} , r_{out} , the control count and the seed are columns of the released per-window catalogue.

The geometry is reconstructed from the code, not the frozen products, so two checks establish that it is the layout that ran. The inner check-ring it generates has exactly the 135 positions the retired region-max statistic maximised over (§5.4). And where extended CO emission fills the beam, the released control statistics fall with the reconstructed radius: Spearman $\rho = -0.49$ at $p \sim 10^{-32}$ in the HD 48370 Band 6 window, against a median $\rho = 0.009$ over the 304 windows containing no crossing at all. A wrong position-to-index mapping could not produce that.

What the geometry costs. A star near the pointing centre sits at primary-beam response ≈ 1 , while the annulus samples 0.95 at 0.14 θ_{PB} falling to 0.19 at 0.78 θ_{PB} , area-weighted mean 0.47 (on ALMA’s actual $1.13 \lambda/D$ beam the same radii give 0.94 to 0.14). Star and controls are therefore not exchangeable in flux density. They are exchangeable in signal-to-noise, what the statistic ranks: primary-beam correction divides an extracted amplitude and its noise

TABLE 2

CONTROL-ANNULUS GEOMETRY BY ALMA BAND, AT EACH BAND'S MEDIAN SEARCHED FREQUENCY. THE ANNULUS IS CONCENTRIC WITH THE STAR, RADII $0.14\text{--}0.78\theta_{\text{PB}}$ WITH $\theta_{\text{PB}} = 1.22\lambda/D$, 512 POSITIONS DRAWN UNIFORM IN AREA AND AZIMUTH FROM SEED 20260825. PER-WINDOW VALUES ARE COLUMNS OF THE RELEASED CATALOGUE; THESE ROWS ARE THE BAND MEDIANS.

Band	ν_{med} (GHz)	θ_{PB} (")	r_{in} (")	r_{out} (")	windows
3	102	61.3	8.6	47.8	40
4	150	41.9	5.9	32.7	4
5	203	31.0	4.3	24.2	4
6	231	27.3	3.8	21.3	216
7	343	18.3	2.6	14.3	155
8	487	12.9	1.8	10.1	12

TABLE 3

THE STATISTICAL DATA PATH, IN ORDER. STEPS 1–8 RUN IDENTICALLY AT THE STELLAR POSITION AND ALL 512 CONTROLS; ONLY STEP 9 ONWARD DISTINGUISHES THEM.

1	calibrated measurement set from the ALMA archive
2	stellar position propagated to the observation epoch (Gaia DR3)
3	spectral extraction at that position, native channelisation
4	per-integration baseline removal: running median along frequency
5	stack along each trial linear drift rate
6	robust per-position noise scale $\sigma(\mathbf{x})$ (MAD)
7	$T(\mathbf{x})$ from Eq. 2
8	repeat 3–7 at 512 control positions on a ring
9	threshold: $T_{\star} \geq 5$ (a crossing)
10	rank test: $T_{\star} > \max_i T_i$ (stage-1 statistical flag)
11	molecular-line mask, $\pm 50 \text{ km s}^{-1}$ stellar frame
12	vetting: recurrence, closure phase, cross-target occupancy

by the same $G(r)$, so $T(\mathbf{x})$ in Eq. 2 is invariant under it. What does not cancel is sky brightness, which is why a beam-filling line raises the ring faster than the star (§5.3). The residual asymmetry is one-sided: a suppressed control can only make a star-exceeds-ring outcome more likely under the null.

Neighbouring controls are correlated below the synthesised-beam scale, so a window's annulus carries $N_{\text{eff}} \simeq 30\text{--}43$ independent spatial trials, as the symmetric reprocessing runs measure (Appendix I). The synthesised beam is not carried by the frozen products, so radii are quoted in arcsec and in θ_{PB} , and N_{eff} is the operative trial count.

Ranking $T_{\star}$ against 512 values fixes the smallest per-window p -value the design can express at $1/513$, while a survey-wide Bonferroni scale over 431 windows is 1.2×10^{-4} , two orders of magnitude smaller. **No single event in this design can therefore reach survey-wide significance.** A stage-1 flag is a feature selected for examination, not a statistically significant candidate. The annulus estimates the local false-alarm rate; it cannot authenticate an individual event (component v, §5.3). Table 3 allows the statistic to be reimplemented without the appendices.

Being global in frequency, that scale can misstate the noise *local* to a crossing wherever a window carries bandpass-edge structure, atmospheric features or a line forest. The retained products keep the per-window scale but not the per-channel spectra, so no local estimate can be recomputed survey-wide; for the four flags we have measured it. Re-extracting their per-integration spectra and recomputing the scale locally, within a band around the feature and excluding the feature itself, gives local-to-global ratios of 0.97 (CP–72 2713), 1.29 (HD 48370), 0.93 (β Pic B3) and 1.02 (β Pic B6). The global scale is accurate to a few per cent except towards HD 48370, where the local value sits 29 per cent higher because that field is filled with CO; rescaling both the statistic and its ensemble by the local value leaves it flagged ($T_{\star} = 21.0$ against a corrected ring maximum of 20.8), the disposition already assigned to it. For CP–72 2713 the local scale is slightly *lower*, so its excess is marginally understated, not inflated, and no disposition moves. The retained products keep no drift-stacked cube, so this tests the noise scale and not $T_{\star}$ itself, and a per-crossing local scale inside the search is an open item (§6.4). The same estimator runs at the star and at all 512 control positions, so a badly misstated σ would displace the control distribution, and the pseudo-star ranks are uniform in every band and at both channelisations (§5.3). The one departing stratum is the fine-channel one, and that departure is the four flagged windows themselves.

Why the statistic is symmetric. The exchangeability that gives a rank its meaning requires the identical operation at the star and at every control; maximising over an n_{src} -position region at the star while comparing against single-position controls breaks that. Under a pure-noise null the region maximum exceeds all n_{ctrl} single controls with probability $n_{\text{src}}/(n_{\text{src}} + n_{\text{ctrl}})$, which for the $n_{\text{src}} = 135$ check-ring and $n_{\text{ctrl}} = 512$ controls is 20.9 per cent, not $1/513$. The inflation is a property of the estimator, holding for any window and any noise distribution, so the released statistic evaluates one position at the star and one at each control, the form of Eq. 2. An earlier version of this analysis used the asymmetric region maximum, and §5.4 states the chronology and audits every window the change affected.

The output vocabulary nests strictly (Table 1).

The searched drift-rate range is the larger of (a) a generic frequency-scaled default ($12\text{--}13 \text{ Hz s}^{-1} \text{ GHz}^{-1}$), the Mason et al. (2024) fiducial with a threefold safety margin, and (b) for confirmed exoplanet hosts, the maximum line-of-sight

orbital acceleration of the innermost known planet, so a transmitter on a short-period planet cannot drift out of the window. Drift rate maps onto line-of-sight acceleration through the first-order Doppler relation

$$a_{\text{los}} = c \frac{\dot{\nu}}{\nu}, \quad (3)$$

so the ceiling is quoted in cross-band-comparable form $\dot{\nu}/\nu$: $12\text{--}13\text{ Hz s}^{-1}\text{ GHz}^{-1}$ is $\dot{\nu}/\nu = (1.2\text{--}1.3) \times 10^{-8}\text{ s}^{-1}$, i.e. $|a_{\text{los}}| = 3.6\text{--}3.9\text{ m s}^{-2}$ (Eq. 3), independent of observing frequency. That is the acceleration at $\sim 0.04\text{ au}$ around a solar-type star. Earth-motion contributions ($\lesssim 0.13\text{ Hz s}^{-1}\text{ GHz}^{-1}$) lie two orders of magnitude below, subsumed by the grid. The grid itself is uniform in $\dot{\nu}$, spanning the searched range in n_{drift} steps (2–1944 across windows, recorded per window in the released file), so the worst-case mismatch, a carrier midway between trial rates, leaves a residual drift that traverses at most 0.26 of one native channel over the longest fine-class track (0.29 over the coarse class), and the stacked-amplitude loss is bounded by that sub-channel smearing alone, an order below the recovery slope measured between 5σ and 10σ . The injection campaign injected on grid points; continuous randomisation between them belongs with the stratified campaign of §6.4. Very short-period systems are the ceiling’s sacrificial edge. An artificial emitter need not be tied to a known planet at all (a free-flying platform, rotating structure or deliberately chirped carrier can accelerate as it chooses), and such emitters are unconstrained, not excluded. These are *apparent* topocentric drift rates, including deliberate transmitted chirp and acceleration-induced drift, which the survey does not distinguish, so the constraints concern linear apparent drift within the searched range only. For circular orbits the ceiling covers a transmitter co-orbiting an Earth analogue by three orders of magnitude and Proxima b comfortably, while TRAPPIST-1 d, c and b ($a_{\text{los}} = 1.07, 2.14$ and 4.00 m s^{-2}) sit at or beyond it. Eccentricity raises the periastron value by up to $(1+e)/(1-e)^2$, $1.9\times$ at $e = 0.2$, so borderline cases move outside the searched domain at modest eccentricity (Fig. 8).

Across the 431 windows maximum searched drift rates span $1.1\text{--}5.9\text{ kHz s}^{-1}$, native channel widths $15.3\text{ kHz--}31.25\text{ MHz}$ (median 15.625 MHz), trial drift rates per window 2–1944 (median 4). The threefold margin matters: at $1\times$ the search would have been blind to two of its threshold-crossing windows, including the AU Mic crossing; at $5\times$ no disposition changes, transmitters drifting faster than $\sim 13\text{ Hz s}^{-1}\text{ GHz}^{-1}$ remaining unconstrained. Intra-integration smearing is negligible almost everywhere: at most 1.10 channels, median 0.001; only three windows have $\eta_{\text{smear}} < 0.99$ (Appendix A), the worst the β Pic Band 6 fine window, whose effective threshold is $1.74\times$ nominal. Channel widths are effective resolutions, verified against the measurement sets’ spectral window tables (Appendix A).

4.2. Frequency and velocity reference frames

Every frequency axis here, searched spectra, tabulated ranges and crossing frequencies alike, is *topocentric* as delivered by the ALMA correlator, with no velocity-frame shift applied at any stage. Velocities enter only in the vetting stage’s line mask, whose conversion chain is in Appendix A. The stacking is coherent in frequency and in the drift-rate trial but preserves no electric-field phase coherence between integrations.

The native channel width is also the linewidth convention for every EIRP threshold, the commonest confusion when technosignature limits are compared across surveys. For a transmitter intrinsically narrower than the channel ($\delta\nu_{\text{tx}} \ll \Delta\nu_{\text{ch}}$), channel dilution spreads its total line flux F_{line} across the channel, $S_{\nu,\text{ch}} = F_{\text{line}}/\Delta\nu_{\text{ch}}$, and triggering requires $S_{\nu,\text{ch}} \geq S_{\text{min}}$, so the minimum detectable total power is $4\pi d^2 S_{\nu,\text{ch}} \Delta\nu_{\text{ch}}$, independent of $\delta\nu_{\text{tx}}$ below the channel width. $\text{EIRP}_{5\sigma}$ is thus the minimum *total* power of a channel-confined carrier and not a spectral power density; dividing by $\Delta\nu_{\text{ch}}$ converts to the minimum detectable spectral power for wider transmitters, so a 10^{15} W threshold at 15.625 MHz channelisation is $6.4 \times 10^7\text{ W Hz}^{-1}$. The 1-Hz and 3-Hz limits of Hz-resolution surveys are total powers in the same sense, each the best case of its own channelisation, which makes the axes of Fig. 1 commensurable. Channelisation is the dominant price the archive exacts, but we do not project an Hz-resolution sensitivity: a radiometer-equation scaling $P_{\text{min}} \propto \sqrt{\Delta\nu}$ across six orders of magnitude in spectral resolution is a thought experiment, not an achievable ALMA sensitivity.

The spectral-carrier search covers every spectral window of an observation, capped at 8 per target to bound compute cost. The cap bound one target/band, 61 Vir Band 7 (four of eight windows searched), whose frequency-space selection function is tabulated. Of the 107 star/band datasets, 53 have every configured window searched; the remaining 3 (τ Cet, Kapteyn’s Star, Van Maanen’s Star) are represented by their deepest single window and are marked in the released catalogue. Each searched window yields its own $\text{EIRP}_{5\sigma}$ row, and Figure 1 plots only the deepest per star.

Velocity coincidence, not drift, is the operative discriminant when a crossing is checked against the masked species: circumstellar emission sits at rest offset by the gas’s kinematics, whereas a technosignature may appear at any frequency and drift rate. The searches as executed used a narrower one-channel-width implementation, which agrees with the velocity-space form on every disposition here (§5.3).

Both carrier-lane thresholds and continuum fluxes are corrected for ALMA’s primary-beam response at the star’s offset from the phase centre, negligibly for most targets (median factor 1.0001, 90th percentile 1.012, none above 2 per cent). Two sit at large offset. ϵ Eri Band 6 lies at ~ 0.7 primary-beam FWHM, where the correction reaches $2.9\text{--}3.4\times$ and the Gaussian beam form fails; rather than publish provisional values we *withhold* those four windows, which enter no number here but are released, flagged, in the catalogue and return once a holography beam model is applied. Sirius B is retained and quoted as such, with corrections of $3.6\text{--}16$ per cent across bands, a bounded systematic, not an invalid model. Excluding it too would change neither endpoint of the quoted EIRP range, nor the median ($1.3 \rightarrow 1.4 \times 10^{15}\text{ W}$), nor any conclusion.

The EIRP thresholds are compared throughout to an Arecibo-like planetary radar (Drake 1974; Sheikh et al. 2025a, EIRP 2×10^{13} W), which is a technological *power* scale and not a matched transmitter model. It is an S-band number, and aperture gain scales as ν^2 , so the same aperture and transmitter power at 230 GHz would radiate $\sim 4 \times 10^{17}$ W. The frequency-matched benchmark is a 12-m aperture radiating 1 MW at 230 GHz, EIRP $\approx 8 \times 10^{14}$ W, within a factor of two of the median Class A threshold 7.5×10^{14} W. Neither is a model.

4.3. Ancillary and validation analyses

Four analyses beyond the spectral-carrier search are executed here, each summarised only as far as the main text depends on it.

Injection-recovery validation (Appendix B). Synthetic tones of known amplitude were injected into real visibilities and recovered with the unmodified pipeline: 1200 trials across six window configurations, three bands and both resolution classes. The pipeline’s baseline step is a per-integration running median along *frequency* (Table 3, step 4), which by construction cannot remove a feature confined to one or two channels, and its de-drift step is an inverse-variance-weighted sum over integrations, which accumulates a persistent carrier coherently. One validation-history item, told in full in Appendix B: the campaign’s original report of 0 of 500 coarse-window recoveries was an artefact of its recovery *criterion*, not of the search pipeline. The corrected criterion has not been re-derived by an independently written implementation, and that cross-check is an open item (§6.4). The independent evidence is the dwell campaign of §4.4, which measures persistent near-static carriers as *recovered* (Table 4), the reading the artefact had reversed.

The primary statistic has no automated injection test on every pipeline change; three quantified checks bound undetected pipeline faults instead: the machinery-only lane (fidelity $-9/+14$ per cent, the bound that caught this artefact), the AU Mic re-analyses of §5.4 under three treatments, whose recomputation agrees with the released products, and the pseudo-star rank uniformity (220 672 pooled ranks uniform, mean 0.5010 against 0.5, KS $p = 0.37$), which any defect of the statistic itself would displace. For the drifting class on the fine window, recovery is 17, 42, 58, 75 and 83 per cent at 4, 5, 6, 8 and 10σ , measured on the single calibration configuration of one star and one Band 6 window and transferred by assumption to all 118 Class A windows, an error the bracketing of Appendix K treats as a $0.5\text{--}2\times$ rescaling. That gives EIRP₅₀ = 1.10 EIRP_{5 σ} and, as a bound, EIRP₉₀ > 2.0 EIRP_{5 σ} , each with an unbounded transfer systematic bracketed as $0.5\text{--}2\times$ (Fig. 7, Table 11). Tables here stay in nominal EIRP_{5 σ} .

4.4. Dwell-fraction completeness

The searched class is defined by how long a carrier occupies one native channel, so we measure recovery against that directly. A carrier was injected into the retained per-integration spectra of 12 real windows, six coarse and six fine, at amplitudes of 2–20 times the per-integration noise and dwell fractions $f_{\text{dwell}} = 0.10\text{--}1.00$, six realisations each: 3888 trials, released with one row per injection (Table 4). Its detection criterion, recovered when the logged per-integration S/N reaches five, was independently re-derived from the released rows: over 3888 trials and 12 windows the rule and the logged flag agree 3888 times, with 0 mismatches, highest unrecovered S/N 5.00 and lowest recovered S/N 5.04, a clean separating gap. Recovery *rises* monotonically with dwell. Pooling amplitudes at or above 4σ , coarse windows recover 100 per cent of continuous carriers, 100 per cent at $f_{\text{dwell}} = 0.25$ and 83 per cent at 0.10; fine windows give 100, 91 and 71 per cent.

An end-to-end check confirms this through the unmodified pipeline, not a re-implementation. Injecting a zero-drift, always-on tone into the calibrated visibilities of a coarse Band 7 window at 1, 3 and 10 times the per-integration rms and running the released extraction and search returns $T_\star = 20.5, 65.3$ and 217.7, each recovered within one channel of the injection and flagged as a detection. For 639 integrations the ideal coherent gain is $\sqrt{639} = 25$, so a carrier at the per-integration noise level should reach $\sim 25\sigma$: the measured 20.5 is that, less weighting and flagging losses.

Three distinctions discipline how these numbers are read. *Injection point*: the dwell campaign injects into retained per-integration spectra, after the baseline step of Table 3, while the end-to-end check injects into visibilities, before it. *Drift discrimination*: on Class B windows every in-grid drift is sub-channel over the track, so the search recovers the carrier’s amplitude but carries no information on its true drift rate, and no drift-rate claim is made for the 313 of 431 windows in that class. *The one cell measured only in part*: through-pipeline recovery of near-static carriers on Class A windows, never yet constrained by an end-to-end zero-drift injection (Appendix B). The earlier claim that the pipeline suppresses persistent carriers is withdrawn.

The campaign measures completeness at the survey threshold, across configurations. The trigger acts on the stacked statistic and the de-drift step is an inverse-variance-weighted sum over integrations, so a carrier of per-integration amplitude A present for a dwell fraction f_{dwell} of a track of N_{int} integrations accumulates to

$$a = A f_{\text{dwell}} \sqrt{N_{\text{int}}} \quad (4)$$

in units of the per-integration noise. Recovery is a function of a , not of A and f_{dwell} separately, so binning the released trials in a unifies the two axes and places all 12 injected windows on one axis. Those windows span $N_{\text{int}} = 21\text{--}1314$, a factor of 63, over 11 targets in Bands 3, 6 and 7 and both resolution classes, so configuration-independence is measurable here, not assumed. In the 6 windows whose amplitude grid straddles the trigger, the 50 per cent recovery point lies at $a = 5.3\text{--}8.3$ (median 6.2), which is the 5σ threshold itself, not a value set by channelisation, integration count or band. In the other 6 the weakest injection already stacks to $a = 3.4$ and at least 97 per cent of every injection is recovered, consistent but uninformative about the curve’s position. The pooled curves (Table 4) agree between classes within the sampling noise.

TABLE 4

THE DWELL CAMPAIGN. (a) RECOVERY OF HIGH PER-INTEGRATION-S/N INJECTED CARRIERS AS A FUNCTION OF DWELL FRACTION, POOLING INJECTED AMPLITUDES $\geq 4\sigma$ PER INTEGRATION, FROM 3888 TRIALS OVER 12 REAL WINDOWS. RECOVERY RISES WITH DWELL: A CONTINUOUSLY TRANSMITTING CARRIER IS THE EASIEST CASE. THIS EXPERIMENT ESTABLISHES THE DEPENDENCE ON TEMPORAL OCCUPANCY, NOT THE ABSOLUTE COMPLETENESS AT THE SURVEY DETECTION THRESHOLD (THAT IS THE END-TO-END CAMPAIGN OF FIG. 7).

f_{dwell}	coarse windows	fine windows
1.00	100%	100%
0.50	100%	99%
0.25	100%	91%
0.10	83%	71%

(b) completeness at the trigger, against stacked amplitude a (Eq. 4), pooled over the 12 injected windows; trial counts in parentheses.

stacked a	Class B (coarse)	Class A (fine)
< 3	0% (30)	0% (72)
3–4	0% (12)	7% (42)
4–5	3% (30)	12% (42)
5–6	46% (24)	33% (48)
6–8	58% (60)	70% (96)
8–12	100% (108)	99% (174)
> 12	100% (1680)	100% (1470)

What this does and does not license. The injected carriers do not drift, so what is measured across 12 configurations is the persistent and partial-dwell, non-drifting class. On Class B windows that is the searched class, which therefore carries a survey-threshold completeness curve measured on six configurations where it had none. On Class A windows the searched class is the drifting carrier, whose curve is still the single calibration configuration of Appendix B; the fine column of Table 4 constrains the near-static cell at the search stage only, and neither replaces the drifting-class curve nor bounds its transfer error. The campaign injects after the baseline step, so it measures the search stage, the visibility-domain injection above confirming the stage before it on one coarse window. And a is a derived combination, justified by the form of the statistic and by the collapse of 12 configurations onto one curve, not measured independently at every (A, f_{dwell}) pair.

Drift-ceiling caveat. The fixed $12 \text{ Hz s}^{-1} \text{ GHz}^{-1}$ searched ceiling, line-of-sight $|a| = 3.6 \text{ m s}^{-2}$, is uniform across every window: two orders of magnitude clear of Earth-rotation and Earth-orbit drifts, but 11 per cent *below* the 4.0 m s^{-2} of a close-in planet on a TRAPPIST-1 b-like orbit. The limits of §6 are therefore conditional on it (Figure 8).

Closure-phase vetting (Appendix C). A one-sided point-source test, unavailable to single-dish and beamformed surveys because it needs multiple independent baselines. It is implemented and simulation-validated, and at this survey’s flux densities it has *no discriminating power*: per-baseline S/N never exceeds 5.1 in the positive-control fields, so no flagged window could have failed it whatever its nature. It is a proof-of-principle diagnostic, not an operative filter, and no window is described as having “survived” it.

Extended and cross-target screens (Appendices C and D). Satellite constellations grew across the 2013–2025 archival baseline. No known ordinary downlink allocation maps into the searched frequencies, but that excludes neither harmonics, intermodulation products nor unconventional emitters, and no exclusion claim rests on allocation alone. The primary defences are the control ensemble and the time-domain tests. A satellite enters star and control extractions alike within an integration, the channel-width mismatch and the drift-stacked statistic dilute transient passes, and a pass confined to one block and one target would evade the cross-target occupancy check by construction. A near-field emitter need not share a celestial source’s far-field interferometric response, one reason a dedicated per-integration RFI screen and visibility-domain checks belong in any successor analysis. A chirped (quadratic-drift) extension and a per-channel periodicity search ran on the 122 retained spectra as a demonstration, not a survey-grade screen; null behaviour is as expected (star-exceeds-ensemble rates 10.7 and 13.1 per cent against an 11-per-cent chance rate), no instance is a candidate and no limit derives from them. Separately, crossings towards different targets were cross-matched for recurrence at the same sky frequency, the archival substitute for the ON/OFF cadence a single-pointing survey lacks. Of the 366 star pairs with overlapping searched ranges, none falls within the matching kernel, subject to one bookkeeping caveat in the appendix.

The pipeline as released differs from earlier working versions in five substantive ways, each settled before the analysis was frozen and logged with its effect on released products in Appendix J and the repository change-log. Asymmetric-derived quantities are historical there and enter no operative number.

Scope of the constraints. *What was searched:* 33 frequency islands totalling 93.1 GHz (88.2 GHz after molecular-line masking) at 15.3 kHz–31.25 MHz channelisation, during the analysed archival intervals, for carriers within the searched drift ceiling and one native channel in width, at nominal trigger thresholds whose measured recovery is 42 per cent (single configuration, transferred by assumption). Inside that box the non-detection is a constraint. *Nothing in this paper bounds the prevalence of:* (i) drift-rate structure on the 313 coarse windows, which carry

TABLE 5

THE SURVEY AS SEARCHED, AND THE PAPER’S HEADLINE NUMBERS IN ONE PLACE. EVERY COUNT QUOTED IN THIS PAPER RECONCILES WITH IT; SUBSET NUMBERS ARE IDENTIFIED WHERE USED. THRESHOLDS ARE NOMINAL (BOXED RULE, §4). THE TWO COMPLETENESS EXPERIMENTS MEASURE DIFFERENT QUANTITIES AND ARE KEPT IN SEPARATE TABLES. THE CONDITIONAL SEARCHED-DOMAIN TRANSMITTER FRACTION IS A METHODOLOGICAL DEMONSTRATION (APPENDIX K) AND IS DELIBERATELY ABSENT.

Quantity	Value
Census stars within 40 pc	168
Stars searched; independent systems	88; 81
Star/band datasets	107
Spectral windows searched	431
extracted; repeats, defective, withheld	462; 21, 6, 4
Class A drift-resolved / Class B unresolved-excess	118 / 313 (73% Class B)
Execution blocks	102
Distance range	1.30–39.63 pc
Unique frequency union	93.1 GHz (33 intervals, 89.6–495.1 GHz)
Windows by ALMA band (3/4/5/6/7/8)	40/4/4/216/155/12
Effective after line-exclusion mask	≈88.2 GHz
Native channel widths	15.3 kHz–31.25 MHz; median 15.625 MHz
Drift-rate ceilings	1.1–5.9 kHz s ^{−1} (12.0–13.3 Hz s ^{−1} GHz ^{−1})
On-source time per window	21–5274 s (median 2087 s)
Nominal $S_{\min}$	0.48 mJy–1.30 Jy (median 6.7 mJy)
Nominal P_{trig} , Class A (15.3–1953 kHz)	2.3×10^{13} – 6.8×10^{16} W (median 7.5×10^{14})
Nominal P_{trig} , Class B (15.625–31.25 MHz)	1.6×10^{13} – 2.8×10^{17} W (median 2.0×10^{15})
worst-case Hanning-corrected medians	2.0×10^{15} / 5.4×10^{15} W
Completeness, by class and experiment	Table 11 (amplitude); Table 4 (dwell)
Threshold crossings ($\geq 5\sigma$ at the stellar position)	20 windows (all Class A)
Star-exceeds-ring windows ($\geq 5\sigma$)	4 windows (all Class A)
circumstellar or foreground CO	3
unclassified	1
Credible technosignatures	0

no drift discrimination and no survey-threshold completeness curve (§4.4); (ii) transmitters emitting *outside the searched spectral windows*, meaning those 33 islands, not ALMA’s frequency range (Fig. 2); (iii) transmitters active *outside the archival epochs*: the limits assume near-continuous emission and are unconstrained below epoch activity ~ 0.1 (Table 13); (iv) drift rates *above* the searched ceiling $|a| = 3.6 \text{ ms}^{-2}$, excluding the most close-in orbits (Fig. 8); (v) carriers placed *inside* the $\pm 50 \text{ km s}^{-1}$ molecular-line mask, which is excluded search space and not searched-and-empty; (vi) emission *broader* than one native channel, for which the thresholds must be rescaled; (vii) anything below the nominal threshold, where measured recovery is 42^{+16}_{-14} per cent (Appendix B); (viii) any transmitter-frequency prior spanning ALMA’s tuning range, under which the conditional bound vanishes (§6.2).

5. RESULTS

The results are presented in three tiers of decreasing evidential weight: the search itself, its non-detection, per-window thresholds and measured completeness (§5.2, §5.3); the four flagged windows and their dispositions (§5.3, §5.4); and the ancillary by-products, continuum, line catalogue and audits, which gate nothing in the primary lane (§5.5). One robustness statement belongs with the headline: the non-detection is stated under two late-stage corrections, both of which made the reported picture more conservative. The mid-analysis statistic revision (§5.4, Table 9) only ever removed flags (seven windows before, four after, none in which the star dominated its own controls), and withdrawing the coarse-window 0-of-500 recovery artefact (§4.3) restored a sensitivity claim (persistent near-static carriers are recovered, not suppressed) rather than removing one. The evidence for both consolidates in Appendix J.

Strict volume ordering (nearest star first; §3) weights these results to the nearest stars, spanning 1.30–39.63 pc. Per-target values, including each band’s searched frequency range, are in the released catalogue (Data Availability).

5.1. Barnard’s Star and Wolf 359

Barnard’s Star and Wolf 359, the second- and fourth-nearest stellar systems, have not to our knowledge been searched for technosignatures at ALMA frequencies before. Both are non-detections in public Band 6 archival data: nominal EIRP_{5 σ} trigger thresholds of 6.2 – 6.9×10^{13} W (Barnard’s Star, 4 windows) and 7.8 – 9.1×10^{13} W (Wolf 359), with continuum upper limits of 0.575 and 0.453 mJy.

5.2. Data-quality exclusions and open items

107 target/bands are searched to completion and reported here. Four classes of entry are absent from the released catalogue and itemised in Appendix J: four process-level search failures, each still carrying a continuum non-detection; one pair excluded for data quality; windows withheld for a noise-handling defect; one relabelled entry.

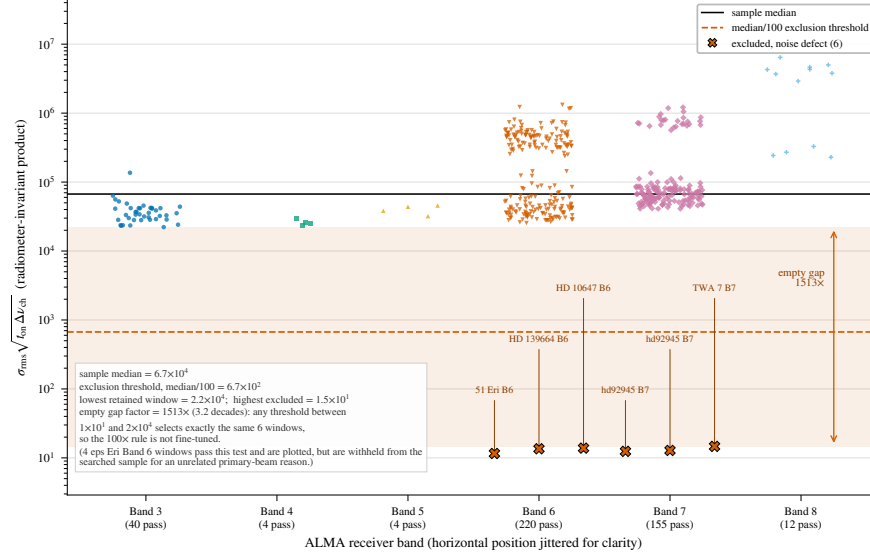


FIG. 3.— Quality control on the noise estimate (§5.2). The plotted $\sigma\sqrt{t_{\text{on}}\Delta\nu_{\text{ch}}}$ is the radiometer equation rearranged: removing integration time and channel width leaves a quantity fixed by system temperature and collecting area, so real ALMA windows cluster within a factor of a few whatever the setup. The six excluded windows sit a factor 1.5×10^3 below the lowest retained one, across an empty gap.

THE COARSE-NOISE DEFECT, AND THE BOUND ON ITS REACH

A reproducible defect produces implausibly small noise in a minority of coarse-channel windows. The radiometer relation $\sigma \simeq \text{SEFD}/\sqrt{N_{\text{bl}}\Delta\nu_{\text{ch}}t_{\text{on}}}$ makes $q \equiv \sigma\sqrt{t_{\text{on}}\Delta\nu_{\text{ch}}} \simeq \text{SEFD}/\sqrt{N_{\text{bl}}}$ a prediction, not a convention, so a window far below the sample median is a bookkeeping defect, not a fluctuation; we exclude such windows at $100\times$ below the median. The failure mode itself is not identified. Six of 431 windows fail, across five stars, and their nominal thresholds of $2.0\text{--}7.8 \times 10^{12}$ W would otherwise have set this paper’s headline depth; excluding them, the best is 1.6×10^{13} W.

The exclusion does not rest on the retained population being narrow, which it is not: recomputed from the released products, q runs from $0.33\times$ to $96\times$ the sample median, a factor of ~ 290 , with a secondary mode from the Band 8 windows (band median $56\times$) and a factor-52 spread within Band 6 alone. The same relation predicts that band dependence through SEFD, and normalising q per band would turn this screen from an outlier cut into a physical comparison. The exclusion survives because the six defective windows sit a factor 1.5×10^3 below the lowest retained one across an empty gap of 3.2 decades (Fig. 3), and any threshold between 1.5×10^1 and 2.2×10^4 selects the same six. The load-bearing argument is one-directional: a window-level deflation of σ acts at stellar and control positions alike, so it can only over-state sensitivity, never manufacture a star-exceeds-ring window, and the same holds for a weaker recurrence hidden in the retained population. What is not known, while the mechanism is unknown, is whether the defect is window-level or position-level, and the check that would settle it, whether the 512 control σ values in a defective window show the stellar one’s deflation, needs per-position noise records the frozen products do not carry.

21 of the 462 extracted rows are duplicate catalogue entries for one physical window, the same (star, execution block, frequency range, channel width) key appearing twice in the archive query. All are repeated archive products or alternative calibrations of one execution block, not independent observations, so one row per key is the right statistical unit. The retention rule is deterministic and quantity-independent, the key’s first-listed row in the archive query’s own identifier order, so it consults no measured value and cannot prefer a crossing or a deeper threshold. Two repeats are not identical: two reductions of the same τ Ceti Band 6 window differ by 44 per cent in threshold, and the two HR 1010 Band 6 rows straddle the 5σ threshold at 5.10 and 4.95, so the crossing count is rule-dependent at the level of one window. The four flagged windows are not, no discarded row being both a crossing and star-exceeds-ring, and every row ships with its status. The audit is otherwise clean apart from four windows of elevated system temperature, and every sample identifier re-resolves in SIMBAD and *Gaia*. Item (iv) is a cautionary example for archival work: the catalogue entry “ γ Lupi” resolved to the F4V debris-disc host HD 139664, holder of the historical identifier “* g Lup”, a name-lookup substitution caught by inverting the pipeline distance to recover the *Gaia* DR3 parallax (Appendix J).

5.3. Technosignature search

The sample is ALMA-archive-complete within 40 pc and not a volume-complete stellar sample (§3). All 107 star/band datasets yielded at least one carrier-lane result (process-level failures, §5.2, uncounted), spanning 431 spectral-window measurements after the catalogue audit removed 21 duplicate rows (Appendix E); B6/216 and B7/155 dominate (per-band breakdown in Table 12). EIRP $_{5\sigma}$ trigger thresholds span 1.6×10^{13} – 2.8×10^{17} W, median 1.4×10^{15} W (Figures 2, 1); as expected for a volume-ordered survey the deepest are the nearest systems’ (1.6×10^{13} W for the UV Ceti pair at 2.7 pc) and the shallowest η Corvi, the first Band 8 target searched. Against the benchmarks of §4.2, only the two resolved components of G 272-61 at 2.7 pc reach below the Arecibo-like planetary-radar EIRP.

The statistics come strictly before the astrophysics, in three named stages that do not promote one another. A

TABLE 6

THE FOUR WINDOWS CONTAINING A STAGE-1 STATISTICAL FLAG, WITH THE EVIDENCE DISPOSITIONING EACH. T_* IS THE SEARCH STATISTIC AT THE STELLAR POSITION (EQ. 2) AND ‘RING MAX’ THE LARGEST OF THAT WINDOW’S 512 CONTROL STATISTICS. ν_{cross} IS THE CROSSING CHANNEL, NOT THE WINDOW CENTRE, THE TWO DIFFERING BY UP TO 0.85 GHz HERE, AND $S_{\text{peak}} = T_*\sigma$ IS THE PEAK FLUX DENSITY IN ONE NATIVE CHANNEL. $\Delta\nu$ AND Δv ARE THE OFFSET OF THAT CHANNEL FROM THE NEAREST *laboratory* REST FREQUENCY OF THE MASKED SPECIES, TOPOCENTRIC, RADIO CONVENTION; THE $\pm 50 \text{ km s}^{-1}$ MASK APPLIES IN THE STELLAR FRAME AFTER THE CONVERSION CHAIN OF APPENDIX A, AND THE REPAIRED MASK OF §5.3 CHANGES NO ROW. THE FROZEN MASK STORED REST FREQUENCIES ROUNDED TO 1 MHz, SO THE RELEASED OFFSET COLUMN READS 0.202 MHz (0.53 km s^{-1}) SMALLER THAN THE VALUES HERE FOR CO(1→0). NO DISPOSITION RESTS ON A SINGLE TEST. THE HD 48370 ROW IS THE LIMITING CASE, A NEAR-TIE OF RANK DISPOSITIONED BY VELOCITY COINCIDENCE WITH INDEPENDENTLY PUBLISHED EMISSION (§5.3).

Window	ν_{cross} (GHz)	T_*	ring max	S_{peak} (mJy)	$\Delta\nu; \Delta v$ (MHz; km s^{-1})	Disposition
β Pic B3	115.260644	14.65	7.27	54	-10.56; -27.5 (CO 1→0)	circumstellar CO
β Pic B6	230.516128	11.68	9.12	78	-21.87; -28.4 (CO 2→1)	circumstellar CO
HD 48370 B6	230.511674	27.10	26.83	0.82×10^3	-26.33; -34.2 (CO 2→1)	foreground CO
CP–72 2713 B7	344.269746	5.81	5.68	12	-1526.24; -1323.2 (CO 3→2)	unclassified; n.s. at survey level

crossing is one channel×drift cell at the stellar position reaching $T \geq 5$: 20 of the 431 windows contain one. A *stage-1 statistical flag* is a window whose stellar statistic also exceeds all 512 controls: four do. That is a screening device and nothing more, selecting a window for examination. The smallest probability it can express is $1/513$, two orders of magnitude above the survey-wide Bonferroni scale 1.2×10^{-4} , so a stage-1 flag is not a statistically significant detection. Each of the four is then carried past that floor by a second-stage local null ((vi) below), which measures one thing: whether a feature exceeds the noise at its own position. It is not a spatial discriminator, and in 3 of the 4 windows the control-ring maximum clears the local null as well, so the first-stage star-versus-control comparison remains the binding test for candidacy. A *candidate* is a stage-1 flag surviving every veto, and there are none. Three of the four flags have independent CO evidence, matched transitions, velocities and epochs at β Pictoris and published foreground emission at HD 48370; the fourth, towards CP–72 2713, is unclassified. The $\pm 50 \text{ km s}^{-1}$ mask is not the instrument of rejection here; the astrophysical evidence is. Table 5 carries the cascade; Table 6 gives each flagged window with the evidence dispositioning it.

None of the 4 stage-1 flags exceeds this survey’s own calibration or is distinguishable from the empirical background or known astrophysical emission.

SURVEY-LEVEL FALSE-ALARM CALIBRATION

Each window-level test compares the peak signal-to-noise ratio at the star’s position with the same statistic at 512 controls. The drift grid and thousands of channels apply identically to both, so the per-window trials factor is absorbed empirically. The statistic is symmetric: the on-source quantity is one propagated stellar position, each control one position. Every number here derives from that frozen statistic; the retired asymmetric variant is development history (Appendix J). The false-alarm probability has five components, in the order the procedure applies them.

(i) *The empirical spatial rank probability.* The per-window significance quantity is an empirical spatial rank probability under the control-position null, not a Gaussian-tail p -value: under exchangeability one designated position of 513 ranks first with probability exactly $1/513$, independent of whether the noise is Gaussian or heavy-tailed, so heavy tails cannot explain an excess. The survey expectation is $431/513 = 0.84$ windows, and $P(\geq 5 | 0.84) = 1.7 \times 10^{-3}$ under an independent-Poisson approximation. That excess over the noise null is real, and astrophysical: in 3 of the 5 windows genuine celestial line emission sits at the stellar position (Table 6), making the star rank first by construction, so those 3 are not draws from the noise null. Removing them leaves 2 rank-first windows against 0.84 expected, $P(\geq 2 | 0.84) = 0.21$: the CP–72 2713 window, and one towards Barnard’s Star whose stellar peak (4.40 against a ring maximum of 4.38) does not reach the trigger and so is not a flagged window. Three levels stay distinct throughout: an instrument and noise null, calibrated by the control ensemble; an astrophysical-source null, addressed by line identification and kinematics; and only then a technosignature reading.

(ii) *The $\geq 5\sigma$ amplitude requirement.* Only 20 of 431 windows contain any $\geq 5\sigma$ crossing at the stellar position, each contributing at most the exchangeable rate $1/513$. The rank, not the trial count, is the calibrated quantity: a window’s channel×drift grid holds up to 1.1×10^5 effective trials (Appendix I), more than 512 controls can resolve, so the control maximum absorbs whatever effective trial count the noise carries. Four flagged windows exceed the budget ($P(\geq 4 | 0.84) = 1.1$ per cent) with no tension: excluding the three astrophysical ones, it need explain one unattributed crossing against 0.81 expected across the 416 non- β Pictoris windows.

(iii) *The line mask.* The velocity-space mask (Appendix F) removes the velocity space of known molecular emission from every crossing before it can count as unattributed. It is an analysis choice, not a measurement: it never lowers the noise-null budget above, and no disposition rests on it alone, the three CO attributions being kinematic (Table 6).

(iv) *The family-level pseudo-target Monte Carlo.* Each trial draws one control per window uniformly from its annulus as the pseudo-star and passes it through the operative gates, $T \geq 5\sigma$ then the velocity mask. The expected count of pseudo-target windows flagged by the full procedure is 0.20 (82 per cent of trials flag none, 18 per cent at least one). This is the prespecified family-level procedure, one expectation carrying threshold, mask and ring geometry together on real data; the Poisson expectation of (i) and the Bonferroni reference of (v) are checks against it, not independent claims. Worked through: 0.84 expected flags from the rank test alone, the amplitude gate and mask removing about

three-quarters of that budget, and the Monte Carlo landing at 0.20.

(v) *The control-ensemble floor.* The controls can show a crossing unremarkable but cannot certify one as significant survey-wide: the smallest expressible per-window probability is $1/513$, two orders above the Bonferroni reference 1.2×10^{-4} over 431 windows, itself conservative because windows share execution blocks. Authentication therefore falls to localisation, recurrence and astrophysical attribution, and we pre-commit that no crossing will be advanced as a candidate on rank statistics alone. No flagged window is promoted by any number in this account.

(vi) *The second-stage local null.* That floor belongs to the first stage, and the four flagged windows are carried past it. For each we take a retained control position, standardise its residual by the pipeline’s per-integration noise map, shift every integration independently and at random in channel, and re-run the identical search: same noise map, weights, drift grid and channel count. A rigid shift preserves each integration’s spectral autocorrelation, 0.65 at lag one, and the maximum runs over the same grid, so the trials factor is the star’s. Independence across integrations leaves nothing coherent along a drift track, so each realisation is signal-free by construction. Between 2×10^4 and 10^5 realisations per window resolve probabilities well below $1/513$. The statistic was reimplemented from the pipeline source and reproduced the published T_* of all four windows bit for bit before any null was drawn.

Each null has first to reproduce the distribution of the 512 real control maxima, genuine draws of the same statistic at exchangeable positions. The criteria were fixed before inspection: 0.2 to 5 per cent of real controls above the null’s 99th percentile, under 1 per cent above its maximum. Three windows pass. HD 48370 Band 6 fails badly: 375 of its 512 real controls, 73 per cent, exceed the maximum of 10^5 null draws, and those controls sit at median 9.30 against the null’s 4.72. The cause is physical, the field carrying extended emission and its imaging residuals, coherent across integrations, which the scramble destroys by design. A scrambled-noise probability there would be manufactured significance, so we quote none for that window.

The two β Pictoris windows act as positive controls: no draw reaches T_* in either, giving $p < 1 \times 10^{-5}$ in Band 3 and $p < 5 \times 10^{-5}$ in Band 6. That is what a working statistic should return where a genuine localised astrophysical signal is present, here on independently known CO. For CP–72 2713 the value is resolved rather than bounded: 157 of 10^5 draws equal or exceed T_* , so $p = 1.6 \times 10^{-3}$. An extreme-value fit to the 512 measured control maxima gives $p = 1.2 \times 10^{-3}$ independently (95 per cent confidence 2×10^{-4} to 3.2×10^{-3}), agreeing to a factor 1.3. Both sit about 13 times above the survey-wide Bonferroni scale 1.2×10^{-4} , and a 2000-fold bootstrap puts the chance of the true value falling below that scale at 1.3 per cent. The same fit places HD 48370 at $p = 1.5 \times 10^{-2}$ (8.3×10^{-3} to 2.2×10^{-2}), unremarkable and consistent with the CO attribution.

Three limits belong with these numbers. A small local-null probability means only that a feature exceeds the noise at its own position, and in 3 of the 4 windows the control-ring maximum clears it too. The retained products keep full dynamic spectra for 8 of the 512 control positions, so position-to-position heterogeneity rests on 8 draws; that is what breaks the HD 48370 null, a limit of the retained products, not of the method, and removable by retaining more control spectra. Parametric extrapolation of a null tail is not used for any number quoted here: for β Pictoris Band 3 the natural fits disagree by 26 orders of magnitude and one implies a bounded tail and returns exactly zero.

Is the ensemble exchangeable with the star? Exchangeability is the central assumption of the statistic: the $1/513$ expectation holds only if the star and its 512 controls are exchangeable under the null, which spatially varying noise, sidelobes, correlated pixels or phase-centre-localised calibration residuals could spoil. The primary-beam term is settled by construction (§4.1): the annulus is concentric with the star and the correction divides signal and noise alike, leaving the ranked statistic invariant. The rest we measure by stratum (Fig. 4). Median per-window control maxima are flat across bands, 4.46, 4.59, 4.42, 4.52, 4.63 and 4.69 in Bands 3 to 8, and a Kolmogorov–Smirnov test of the star’s rank among its controls against $U(0, 1)$ is consistent in every band ($p = 0.20$ – 0.83) and in the coarse stratum ($p = 0.65$). Fine-channel windows sit higher (control maximum 5.76 against 4.42), as their far larger channel $\times$ drift trial count predicts; the test is conditional on the window, so that is no violation.

For the stronger question, exchangeability of the ensemble itself, we ranked each of a window’s 512 controls against the other 512-1 by the statistic and add-one rule used for the star, pooled over the survey. The 220 672 pseudo-star ranks are uniform to the resolution the ensemble can express (mean 0.5010 against 0.5, $p = 0.37$) in every band and at both channelisations. Two caveats belong with it: the construction compares a control against the other 511 plus add-one, a different geometry from star-against-ring, so it cannot detect a centre-versus-annulus asymmetry; and the 220 672 pooled ranks are not independent, 512 coming from each window’s single noise realisation with $N_{\text{eff}} \simeq 30$ – 43 , so the nominal pooled p -values are optimistic. The relevant check is the two-sample comparison of the 431 real stellar ranks against the pooled pseudo-ranks, $p = 0.38$: the stellar position behaves like one more position in its own annulus, and the medians coincide (2.89 against 2.89, $D = 0.030$). In *independent* spatial trials the ensemble delivers only ~ 30 – 43 , so the achievable resolution on a genuinely independent null is nearer a few $\times 10^{-2}$ than $1/513$.

One stratum departs, and is reported, not pooled away. The fine-channel windows fail the same $U(0, 1)$ test at $p = 0.015$ ($n = 118$): four contain an on-star peak above their own control maximum, against 0.22 expected, precisely the four flagged windows of Table 6. That fits the astrophysical line features identified below, not a misbehaving ensemble, but the departure is genuine, so fine-window rank probabilities are slightly optimistic, and localised: excluding the 20 fine windows with any $\geq 5\sigma$ stellar crossing restores uniformity ($p = 0.39$, $n = 98$). Per-control noise records and gain are two stratifications the frozen products cannot support, so the empirical rank distributions bound the net effect: expected flags 0.20 against a rank-only budget of 0.84, with no excess in any stratum. Nor are the windows nearest the beam edge silently affected: Sirius B sits at $r \simeq 0.23 \theta_{\text{PB}}$, inside the annulus’s own radial span, and the four ϵ Eri Band 6 windows are withheld outright (§4.2). Controls at equal holography-model gain would remove the

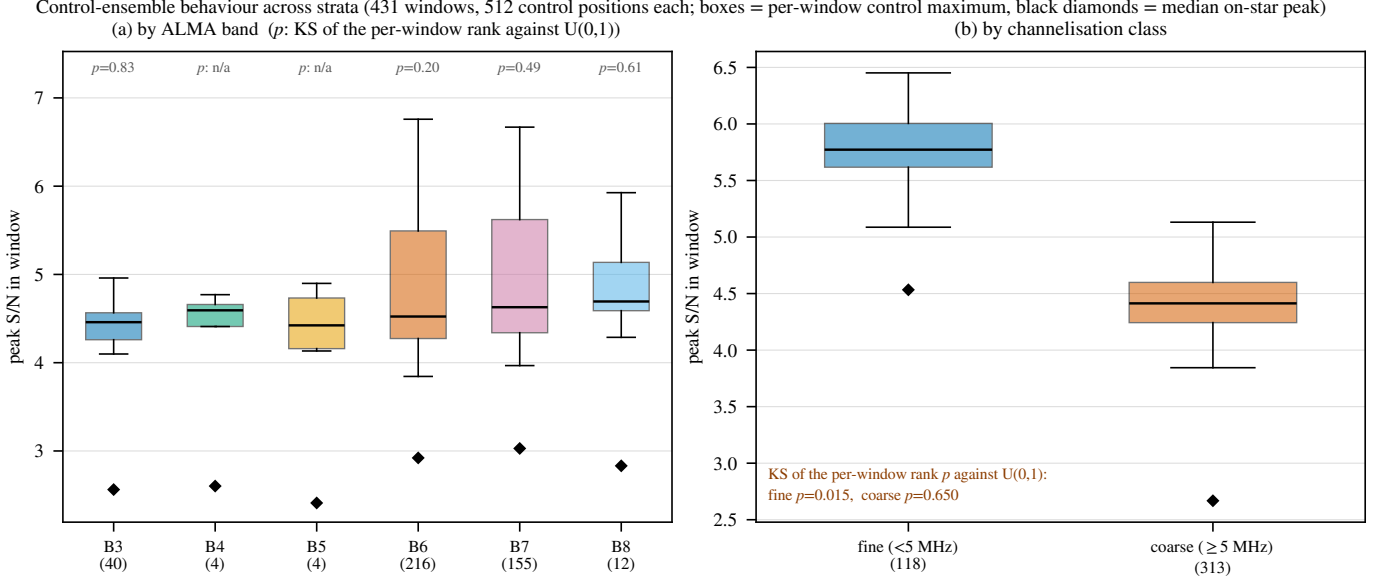


FIG. 4.— Exchangeability of the 512-position control ensemble by stratum. Boxes: per-window control maximum; diamonds: median on-star peak; p : the Kolmogorov–Smirnov test of the text. The ensemble tracks the on-star statistic in every band and both channelisation classes. The one departure is the drift-search class ($p = 0.015$), driven by the four flagged windows of Table 6.

sky-brightness term too, the natural next step. Remaining robustness checks, trials-count reconciliation under channel adjacency above all, are in Appendix I.

THE FULLY SYMMETRIC REGION-MAX CHECK

The SR statistic applies identical source-region, drift-rate and frequency maximisation at the star and at each of the 512 controls under one search rule. The frozen release products cannot meet that, so its domain is the seven locally reprocessed windows: six validation windows plus the AU Mic window’s first-epoch block. All give $T_{\star} = 3.7\text{--}4.9$ against control maxima of $6.7\text{--}7.7$, $p = 0.81\text{--}1.00$, so the star is exchangeable with its ring wherever the test runs at full symmetry. The single-position (SP) calibration above is therefore the operative release-wide criterion over all 431 windows, the SR statistic a validation subset; the two are never interchanged. The machinery ships with the release, and the full statement is in Appendix I.

ASTROPHYSICAL END-TO-END VALIDATION: β PICTORIS CO

Two of the four flagged windows are towards β Pictoris, a debris disc with resolved CO emission (Matrà et al. 2017; Dent et al. 2014): one in Band 3 at 115.260644 GHz, -10.56 MHz from the CO($J=1\rightarrow0$) rest frequency (115.271202 GHz, Pickett et al. 1998), the other in Band 6 at 230.516128 GHz, -21.87 MHz from CO($J=2\rightarrow1$). Neither coincidence is exact, so the automated one-channel check missed both. Through the frame chain (Table 8) the offsets are one kinematic component, -2.72 and -3.45 km s^{-1} from stellar systemic and 0.73 km s^{-1} apart, in the two lowest CO rotational transitions of a system with detected circumstellar CO. The unflagged lower-resolution Band 6 window, a third epoch six years earlier, independently shows CO at -3.62 km s^{-1} in the same frame: all three epochs and both transitions lie within 0.9 km s^{-1} , deep inside the ± 50 km s^{-1} mask. The Band 6 window carries 114 formal $\geq 5\sigma$ crossings where a channel-confined carrier admits one, and the retained products do not record whether those channels are contiguous, so we quote the count, not a line width. Peak flux densities of 54 and 78 mJy per channel, ratio 1.4, are the right order for this belt’s spatially integrated CO but well below the optically thin LTE ratio ($\simeq 4$), as resolved-out flux in the finer configuration would predict; the comparison with calibrated imaging is indicative, the two weighting the emission differently.

We attribute both crossings to this kinematically offset CO. The pair is the survey’s astrophysical end-to-end validation, a known signal run through the same search as the null results, and it recovers the whole check list: the frequencies to the frame chain’s stated tolerance, the velocity concordance across epochs and transitions, the localisation (the Band 3 stellar peak exceeding all 431 control maxima and the Band 6 peak all but its own CO-filled ensemble), and the drift class, zero to within the grid. Nothing about it was tuned. A three-panel figure of both features and their velocity alignment would complete the record, but the frozen release does not carry the channel-level spectral products of those execution blocks, so the validation rests on the velocity chain of Table 8 and the numbers above.

The attribution is no dismissal, their empirical significance against the survey background being extreme: the Band 3 region peak of 20.3 exceeds all 431 per-window control maxima (add-one $p < 0.0044$) and the Band 6 peak of 12.8 all but one, that window’s own CO-filled ensemble reaching 15.2, the largest here ($p \approx 0.0088$). In robust units, the stellar statistic minus the ring median over $1.4826\times$ the ring median absolute deviation, the standardised excesses are $\zeta = 36.0$ and 26.8 , the yardsticks for the remaining flagged windows. The alternative needs a transmitter coincidentally placed at matched offsets from two CO transitions in the same star.

DISPOSITIONS OF THE REMAINING FLAGGED WINDOWS

The third, towards HD 48370 in Band 6, is the largest excess here in absolute and the smallest in relative terms: $T_\star = 27.10$ against a control-ring maximum of 26.83, the star exceeding its ring by one per cent of the statistic. Its standardised excess, $\zeta = 4.2$, is near the CP-72 value below, though the amplitude and velocity evidence differ. Star and ring rising together is what emission filling the primary beam looks like, and the window sits -34.2 km s^{-1} topocentric from CO($J=2 \rightarrow 1$). The attribution is independently published: Cataldi et al. (2023) report prominent CO($2 \rightarrow 1$) and $^{13}\text{CO}(2 \rightarrow 1)$ emission towards HD 48370 at $v_{\text{bary}} \approx 41 \text{ km s}^{-1}$, well separated from its 23.6 km s^{-1} systemic velocity and attributed by them to foreground cloud contamination; our window recovers it, and the crossing is consistent with, and very likely caused by, it. Two internal corroborations make this the most secure, not the most fragile, disposition. The peak is 0.82 Jy in one 122-kHz channel with the ring at the same level, a bright line filling the beam and not a marginal statistical excess. And the $^{13}\text{CO}(2 \rightarrow 1)$ window of the same execution block has a control maximum of 20.4 against only 4.6 at the star, so there is bright ^{13}CO in the field but not at the star, as extended foreground emission predicts. The reconstructed annulus geometry (§4.1) locates that peak: the control carrying the maximum lies 9.2 arcsec from the star, $0.32 \theta_{\text{PB}}$, and the ten strongest controls all fall between 7.4 and 11.1 arcsec against a median control radius of 15.9 arcsec: emission offset and clustered, not centred on the star. A third window is consistent: all twelve Band 8 windows here belong to η Crv, HD 48370 and HD 61005 and are [C I] $^3P_1 - ^3P_0$ (492.160651 GHz) tunings, and HD 48370's carries a 5.05σ on-star crossing against a ring maximum of 6.23. Its crossing channel sits -129 km s^{-1} from [C I], too far for the repaired mask to exclude it (§5.3), so its control ensemble disposes it, not line identification. The coincidence is with Cataldi et al.'s quoted v_{bary} , not an independent frame-chain re-derivation on our side, and this window is identified by that velocity coincidence and the near-equality of star and ring, not by the rank statistic: a stated limitation of the spatial test.

THE UNIDENTIFIED CP-72 2713 FEATURE

The fourth flag, towards CP-72 2713 in Band 7, is the only one with no astrophysical attribution, and it is marginal. The star is not anonymous, and the prior for an astrophysical explanation at it is higher than at a random sample member: a K7/M0 member of the $\sim 24 \text{ Myr}$ β Pictoris moving group hosting a cold, dust-rich debris disc detected in ALMA 1.33-mm continuum (Moór et al. 2020), the same population and class of magnetically active star as AU Mic.

What its own data show. The feature is at 344.269746 GHz, not the window centre 345.1152 GHz quoted in earlier versions of this paper: the crossing sits $\sim 0.85 \text{ GHz}$ below it, 35 channels above the window's lower edge, where a global noise scale is least reliable (§4.1). The local recomputation gives 0.97 here, so the excess is if anything understated. Peak flux density at the crossing is 12 mJy in one 488-kHz channel of a 1.73-GHz window, with two adjacent channels crossing threshold. On the numbers, $T_\star = 5.81$ against a ring maximum of 5.68, a margin of 0.13: the stellar position ranks first among the 513 tested positions, but 0.13 is the smallest margin 512 controls can express, a resolution floor and not a measurement of significance. Its standardised excess is $\zeta = 4.6$ in robust units, two orders below the β Pictoris controls, and the margin is smaller than the scatter across this star's own windows, whose other fine-channel control maxima are 5.16, 5.41, 5.67 and 5.85: its ensembles reach 5.81 routinely where the star does not cross threshold. Figure 5 shows the crossing inside its window's full 512-position control distribution, at the upper edge of an unremarkable ensemble, not beyond it, one draw of 513 against expected ring maxima of median 4.56 and 90th percentile 5.85; a ring maximum at or above 5.68 occurs in 17 per cent of windows and at or above 5.81 in 13 per cent.

Two of its own windows bear on the astrophysics. The flagged Band 7 window covers CO($3 \rightarrow 2$) at the stellar velocity and the Band 6 fine window CO($2 \rightarrow 1$); neither crosses threshold, so we report a non-detection of circumstellar CO in the CP-72 2713 belt. That interests the debris-disc community and removes the obvious circumstellar reading of the flag. The nearest transition of the masked species lies $-1323.2 \text{ km s}^{-1}$ away (radio convention; relativistic, -1326), and because that mask is an automated contamination cut, not a complete line catalogue, we re-queried a wider one, whose closest entry is $\text{H}^{13}\text{CN}(J=4 \rightarrow 3)$ at -195 km s^{-1} , still far outside any circumstellar tolerance. The repaired mask of §5.3 changes nothing here. The defensible reading is “no plausible known transition identified”, a statement about the catalogues consulted and not a proof that none exists. Atmospheric or calibration structure near 344.269746 GHz is controlled as everything here is: it enters star and control positions alike within the one execution block, so it widens the ensemble rather than mimicking a stellar excess; isolating any telluric contribution needs the calibration products beneath the frozen release. With a single execution block at this frequency no recurrence test is available.

What is not done, and why. Table 7 lists every quantity the retained products carry. Polarisation hands, time and baseline splits, visibility-domain localisation, calibrator comparison and synthesised-beam localisation all need re-extraction beneath the frozen products. We do not claim tests we have not run. A Gaia/2MASS/WISE cross-match for an unresolved background source is likewise open: its catalogue half could list sources in the synthesised beam, but without visibility-domain localisation it could not say which, if any, the excess belongs to, so a null match would not retire the flag and a populated match could not promote it.

We record the window as an unclassified stage-1 flag, a statistical or astrophysical feature and not candidate evidence, with explicit promote and retire criteria. The second-stage local null replaces the $1/513$ rank resolution with a measured probability: $p = 1.6 \times 10^{-3}$ from 10^5 draws on this window's retained dynamic spectrum, and $p = 1.2 \times 10^{-3}$ from an independent extreme-value fit to its 512 control maxima, both about 13 times above the survey-wide Bonferroni scale. This window's conservative disposition is therefore measured, not asserted. *Promote* only on recurrence at an independent epoch, which a targeted ALMA re-observation at 344.269746 GHz would test directly. No such observation is scheduled or proposed. An enlarged control ensemble is no longer the route it was: the extreme-value fit already

TABLE 7

EVERY QUANTITY THE FROZEN RELEASE PRODUCTS CARRY FOR THE UNIDENTIFIED CP-72 2713 BAND 7 FEATURE. EXECUTION-BLOCK UID, EPOCH AND PROPOSAL CODE ARE IN THE MACHINE-READABLE RELEASE. THE TESTS NEEDING VISIBILITY-LEVEL RE-EXTRACTION OR A TARGETED RE-OBSERVATION ARE LISTED IN THE TEXT; NONE IS PERFORMABLE ON THE FROZEN PRODUCTS.

Quantity	Value	Quantity	Value
Band; centre	7; 345.1152 GHz (bandwidth 1.73 GHz)	Crossing frequency	344.269746 GHz (channel 35 of the window)
Channel width	488 kHz (0.42 km s^{-1})	On-source time	50 min (2994 s); one execution block
Drift grid	129 trials to $\pm 4.14 \text{ kHz s}^{-1}$	Map rms; 5σ	2.10 mJy; 10.5 mJy
T_*	5.81	Ring maximum	5.68 over 512 controls (margin 0.13: the 1/513 floor)
Feature flux	12.2 mJy	Trigger power; EIRP	$8.26 \times 10^{14} \text{ W}$; $9.60 \times 10^{14} \text{ W}$
Stellar distance	36.7 pc	Nearest transition	CO($J=3 \rightarrow 2$), -1326 km s^{-1} away

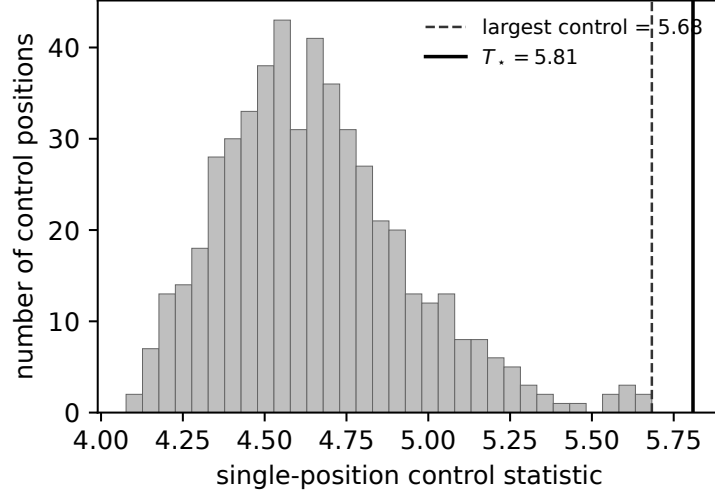


FIG. 5.— The CP-72 2713 Band 7 crossing inside its own control ensemble: histogram of the 512 single-position control statistics of that window, with $T_* = 5.81$ (solid) and the largest control value, 5.68 (dashed), marked.

TABLE 8

LINE-ATTRIBUTION AUDIT OF THE THREE LINE-ATTRIBUTED WINDOWS (FIRST THREE ROWS) AND THE COARSE β PICTORIS BAND 6 WINDOW THAT CARRIES NO STAGE-1 FLAG YET SHOWS THE SAME CO COMPONENT, THROUGH THE FRAME CHAIN OF §4.2. TOPOCENTRIC OFFSETS ARE OBSERVED MINUS CATALOGUED REST FREQUENCY, RADIO CONVENTION (CO $J=1 \rightarrow 0$ 115.271202 GHz, CO $J=2 \rightarrow 1$ 230.538000 GHz, PICKETT ET AL. 1998; THE FROZEN MASK'S OWN ENTRIES, ROUNDED TO 1 MHz, ARE NOT USED HERE); THE BARYCENTRIC CORRECTION IS THE EARTH-EPOCH TERM AT THE EXECUTION-BLOCK MID-TIME (BLOCK UIDS IN THE MACHINE-READABLE RELEASE); SYSTEMIC RADIAL VELOCITIES ARE CATALOGUED (SIMBAD; *Gaia* DR3 FOR β PICTORIS, GAIA COLLABORATION ET AL. 2022; DR2 FOR AU MIC, FOUQUÉ ET AL. 2018).

Flag	UTC start	f_{obs} (GHz)	Transition	Topo. offset (MHz; km s^{-1})	Bary. corr. (km s^{-1})	v_{sys} (km s^{-1})	Stellar-frame offset (km s^{-1})
β Pic B3	2022-03-02 23:14	115.260644	CO($1 \rightarrow 0$)	-10.56; -27.46	-7.90	+16.84	-2.72
β Pic B6	2019-03-12 00:14	230.516128	CO($2 \rightarrow 1$)	-21.87; -28.44	-8.15	+16.84	-3.45
AU Mic B6	2014-08-18 04:01	230.825230	CO($2 \rightarrow 1$)	+287.23; +373.51	-10.02	-4.71	+378.82
β Pic B6 [†]	2013-10-06 06:23	230.528134	CO($2 \rightarrow 1$)	-9.87; -12.83	+7.63	+16.84	-3.62

[†] Coarse Band 6 window without a stage-1 flag (Appendix E), shown because it independently shows the same CO component as that band's flagged fine window.

estimates where a larger ensemble would land, above 1.2×10^{-4} , so the excess would have to grow, not just be resolved more finely. *Retire* on a clean second epoch or on identification with catalogued emission. Until either, it stays released and reported for what it is: unclassified, and not statistically significant at the survey level.

LINE EXCLUSION: DEFINITION, REPAIR, AND COST

The searches as executed used an automated line check matching crossings to catalogued rest frequencies within one spectral channel at zero drift. The β Pictoris windows showed that inadequate: both fell outside its one-channel tolerance, by 10.6 and 21.9 MHz, while lying, after the frame chain, within 3.6 km s^{-1} of the stellar systemic of a resolved, mapped disc, the regime a channel-width tolerance is blind to. Every crossing was therefore reclassified with a velocity-space mask, $\pm 50 \text{ km s}^{-1}$ in the stellar frame around each catalogued transition. Its kinematic justification, empirical conservatism and the counter-argument that a communicator might deliberately choose these natural ref-

erence frequencies are in Appendix F. The mask is applied *retrospectively*, as a post-hoc classification criterion, not a detection threshold: the half-width was adopted after the β Pictoris crossings had been seen. A sensitivity test reclassifying all 20 crossings at ± 20 , ± 30 , ± 50 and ± 100 km s $^{-1}$ masks 2.1, 3.1, 5.2 and 10 per cent of gross bandwidth and admits no new unattributed flag; at every width the only unmasked unattributed flag is CP-72 2713. Masked regions are reported throughout as *excluded search space* and never as searched-and-empty.

The mask as executed had four defects; all four are repaired here. It carried rest frequencies rounded to 1 MHz, worth up to 1.59 km s $^{-1}$; no atomic carbon, although every Band 8 window here is a [C I] 3P_1 - 3P_0 tuning; no hydrogen recombination line, although H30 α falls in the standard Band 6 tuning of much of the sample; and it was applied in the stellar frame only, the wrong frame for the Galactic foreground emission behind one of the three line attributions. The repaired catalogue holds 17 transitions of 11 species at full JPL/CDMS precision (Pickett et al. 1998; Müller et al. 2005), including [C I] at 492.160651 GHz and H30 α at 231.900928 GHz, against 15 of 9 before, and it is applied in the LSR frame as well as the stellar frame. The species count also reconciles the text with the products, which always used 9 species: earlier versions of this paper said eight, omitting H₂CO.

Repairing a mask is cheap; re-running the survey is not. The repaired mask is therefore re-applied to the frozen crossing list as a post-hoc filter, using each crossing channel’s own frequency as the pipeline recorded it. **No disposition changes.** The two β Pictoris crossings remain inside the stellar-frame tube, at -2.72 and -3.45 km s $^{-1}$ from CO(1 $\rightarrow$ 0) and CO(2 $\rightarrow$ 1). HD 48370’s crossing remains inside it at -17.8 km s $^{-1}$ and is now caught by the LSR-frame tube as well, at -23.9 km s $^{-1}$, which is the frame its foreground attribution always implied. CP-72 2713’s crossing lies -1323.2 km s $^{-1}$ from the nearest transition in the topocentric frame and -1300 km s $^{-1}$ in the LSR frame, so no tube in any frame excludes it and it stays unclassified. Across all 20 crossing windows the repaired catalogue moves the nearest transition in exactly 1 case, HD 48370’s Band 8 crossing at 491.949074 GHz, whose nearest entry changes from CO(4 $\rightarrow$ 3) 20098 km s $^{-1}$ away to [C I] at -129 km s $^{-1}$. That is still outside the tube, and the frame corrections are at most a few tens of km s $^{-1}$, so no frame masks it either; it was never a flag, its own control ensemble exceeding it. The two new transitions cost a further 0.24 GHz of unique searched frequency, 0.3 per cent.

The masking cost uses one definition: every JPL catalogued transition of the 9 frozen species (276), displaced by each system’s radial velocity, ± 50 km s $^{-1}$ stellar-frame half-width, intersected with each window and merged. On the unique-frequency union, the survey holds 93.1 GHz of archival spectral coverage, of which 88.2 GHz enters the primary technosignature test and 4.96 GHz (5.3 per cent) is deliberately excluded because of molecular-line confusion (Figure 2). On the window-summed gross bandwidth the same mask removes 22.20 GHz of 716.7 GHz (3.1 per cent), the per-window usable fraction having median 0.960 (10th percentile 0.790). CN and CO dominate the loss, and the per-species, per-band breakdown ships in the released veto-mask export. The exposure accounting is as follows: gross bandwidth 716.7 GHz (Class A 23 per cent, Class B 77 per cent), unique sky-frequency union 93.1 GHz (Class A 29.4 GHz), summed per-star union 371.9 GHz, and on-source $\times$ bandwidth 1.4×10^6 s GHz, or 393 star-hour-GHz. The frequency-exposure curve $C(\nu) = \sum_i t_i(\nu)$ has covered-bin median 2.4 and peak 18 star-hours in 2-GHz bins, and the archive’s tuning concentrations put 30 per cent of all exposure in the five best-populated bins, which any frequency-prior calculation must weight. The mask governs dispositions, not the noise-derived thresholds, so the quoted EIRP $_{5\sigma}$ values are unchanged; no signal was found outside the excluded velocity regions in any window.

The mask is a design selection effect: a signal coincident with a molecular line is not thereby “false” but *excluded from the technosignature search domain*, the astrophysical prior there being overwhelming and this pipeline unable to separate the two. We place no constraint on transmitters in the excluded regions, 5.3 per cent of the unique-frequency union. Masked channels are retained, not discarded, for a line-coincident search with independent discrimination. Nor is the exposure confined to the flagged windows: 131 of 431 windows (30 per cent) contain an in-band catalogued transition of the masked species, as do all four flagged windows, yet in none does a crossing fall within one native channel of a rest frequency, the regime the velocity-space criterion covers.

5.4. Changing the detection statistic, and the AU Mic crossing

The estimator is justified on its own terms in §4.1, from the exchangeability the rank test needs and independently of any dataset. The chronology is this. An earlier region-max statistic maximised over an $n_{\text{src}} = 135$ -position region centred on the star against 512 single-position controls, and under it a crossing towards AU Mic in Band 6 at 230.83 GHz was carried as a candidate; the asymmetry was then identified algebraically, the symmetric form of Eq. 2 substituted, the whole analysis re-run and the criterion frozen; only afterwards was any flagged window, AU Mic’s included, inspected under the new criterion. We state that plainly because the revision removed the one candidate it was not designed to remove, and a reader is entitled to weigh the sequence.

The sequence can be checked against Table 9, which audits every window the original statistic flagged, not the AU Mic window alone. The symmetric criterion flags a strict *subset*, seven before and four after. Both β Pic windows and HD 48370 keep their flags under either statistic; AU Mic, ALMA J1537-3319 and HD 14055 drop out, in all three cases because the stellar position never exceeded its own control ensemble, the one configuration a localised transmitter cannot produce. Under a decision rule fixed without reference to these data, the 1/513 rank test, AU Mic fails on the same evidence: 10 of its 512 controls reach or exceed the stellar statistic. Because the statistic was settled on these data, the false-alarm calibration here is descriptive of them.

The within-window add-one rank is $p = 0.021$, a different quantity from the survey-wide rate at which any window’s ring maximum reaches 5.97 ($p = 0.081$, Appendix G), and far from the 1/513 floor either way. The crossing also fails two pre-named tests, localisation and recurrence. Under the three treatments that decide it, the frozen region-

TABLE 9

EVERY WINDOW FLAGGED BY THE ORIGINAL REGION-MAX STATISTIC, UNDER BOTH STATISTICS, SO THE REVISION CAN BE AUDITED ACROSS ALL OF THEM, NOT JUST THE AU MIC WINDOW. S_{reg} : REGION MAXIMUM OF EQ. 2 OVER $n_{\text{src}} = 135$ POSITIONS; T_* : THE SYMMETRIC SINGLE-POSITION STATISTIC; ‘CTRL MAX’, ‘RING MAX’: THE LARGEST CONTROL STATISTIC UNDER EACH. FOR CP-72 2713 THE REGION MAXIMUM ALREADY LAY ON THE STAR, SO THE TWO COINCIDE.

Star	Band	ν (GHz)	region-max statistic		symmetric statistic		Symmetric outcome
			S_{reg}	ctrl max	T_*	ring max	
HD 48370	6	230.7305	27.95	26.83	27.10	26.83	flagged
β Pic	3	115.2605	20.31	7.27	14.65	7.27	flagged
β Pic	6	230.5164	12.78	9.12	11.68	9.12	flagged
AU Mic	6	230.5384	5.97	5.85	5.22	5.85	not flagged
ALMA J1537-3319	6	217.0183	5.83	5.63	5.09	5.63	not flagged
CP-72 2713	7	345.1152	5.81	5.68	5.81	5.68	flagged
HD 14055	7	345.1377	5.80	5.74	5.22	5.74	not flagged

max products, a fully symmetric reprocessing of the same first-epoch block and the second-epoch block of the same programme, the star statistic is 5.97, 4.93 and 4.77 against control maxima of 5.85, 7.70 and 6.78, with add-one star ranks $p = 0.81$ and 0.97 in the latter two.

Appendix G gives both in full: neither the reprocessed first epoch nor the one further public Band 6 block covering 230.83 GHz shows a localised excess, the second epoch a clean non-recurrence. That rejects a continuous or repeating source at this frequency and these epochs, though not intermittent transmission. Each star was observed in one to three execution blocks, searched independently, so the non-detection concerns the epochs observed, not permanent activity; §6 turns the epoch structure into an explicit epoch-activity constraint.

5.5. Ancillary results

Three ancillary results are reported in full in the appendices.

Integration-time audit (machine-readable release). Window-level on-source records for 227 of the 431 windows were checked against the ALMA archive’s own exposure accounting. Two of the 58 star-execution-block groups over-count, AT Mic by a factor 1.43 and CD-38 10980 by 7 per cent; the rest are conservative. No EIRP threshold depends on integration time, each deriving from the measured per-channel rms, so no disposition changes and the exposure-weighted coverage of §5.3 is at worst optimistic by under 1 per cent.

Continuum search (Appendix H). An exploratory by-product over the 57 target/bands processed far enough for a usable measurement: seventeen $\geq 5\sigma$ detections and forty non-detections reaching 0.036–3.6 mJy (median 0.52 mJy). Every detection is astrophysically unsurprising, the auroral emitter LSR J1835+3259 meriting deeper follow-up. No technosignature disposition rests on it.

Serendipitous molecular-line catalogue (Appendix H). The line-exclusion step identifies every $\geq 5\sigma$ outlier channel as a by-product; across the subset scanned, twelve outlier groups are single- or few-channel features with no known transition within 300 MHz and six are plausibly real, all reported openly.

6. DISCUSSION

6.1. What signals could ALMA archival data actually detect?

Three cases, one per measured morphology, with real windows and the completeness entries of Table 11 (Fig. 6 maps every window). Denominators count the 81 systems whose deepest threshold in that class lies at or below the case power.

(i) *A continuous, frequency-stationary carrier* in a coarse channel is the best case, not the worst. Toward the UV Ceti pair (2.7 pc, deepest window $\text{EIRP}_{5\sigma} = 1.6 \times 10^{13}$ W) such a carrier confined to its channel is recovered at near-ideal $\sqrt{N}$ gain (§4.4), so a carrier matching the *total* EIRP of an Arecibo-like planetary radar (§4.2) is detectable toward the nearest systems at 15.625-MHz resolution. 1 system of 81 qualifies, 2 at twice the power.

(ii) *A continuously drifting carrier* in a fine 488-kHz channel toward τ Ceti (3.65 pc, threshold 3.8×10^{13} W) is recovered 42 per cent of the time at threshold for $f_{\text{drift}} \geq 0.5$ and 75–83 per cent at 1.6 – $2\times$ threshold (Appendix B). A drifting transmitter above $\sim 8 \times 10^{13}$ W is recovered in 83 per cent of trials if its drift stays inside the searched ceiling; 8 systems reach that depth in Class A.

(iii) *An intermittent carrier, on for a quarter of the track*, toward HD 10647 (17.4 pc, 1.5×10^{14} W) is recovered 91 per cent of the time in fine channels, 100 per cent in coarse ones, so a duty-1/4 transmitter above $\sim 2 \times 10^{14}$ W is detected toward 24 such mid-distance systems. Recovery falls rapidly at smaller dwell fractions (Table 4), and below $f_{\text{dwell}} \sim 0.1$ only the nearest windows carry information.

6.2. What does the null result mean?

No credible technosignature has been found (§5.3). The $\text{EIRP}_{5\sigma}$ values of the released catalogue are nominal trigger powers (boxed rule, §4); completeness-calibrated exclusion holds only where the measured recovery curve applies, in Class A. The sample is ALMA-archive-complete within 40 pc and sensitivity-ranked, not an unbiased draw.

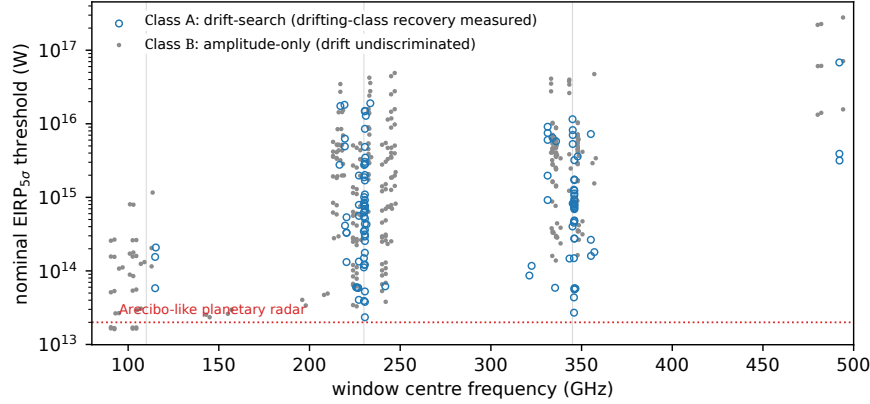


FIG. 6.— Two-dimensional sensitivity: nominal EIRP trigger threshold (boxed rule, §4) against window centre frequency for all 431 searched windows, symbol by channelisation class, the axis along which completeness differs (Table 11). Grey lines mark the Bands 3, 6 and 7 concentrations; dotted, Arecibo-class power.

One exclusion deserves the converse emphasis. The line mask removes $\pm 50 \text{ km s}^{-1}$ stellar-frame tubes around the catalogued transitions, whose rest frequencies are the ‘magic’ channels Cocconi & Morrison (1959) argued a communicator might choose. This survey is blind by construction to a transmitter parked there in the stellar frame, a false-positive control purchased with 3.1 per cent of gross bandwidth and one physically motivated class of signal. Of the 41 planets known around the 18 hosts, only TRAPPIST-1b reaches the drift ceiling in its edge-on worst case, on star–planet relative accelerations GM_*/r^2 from NASA Exoplanet Archive parameters accessed 2026 September 9, maximised over orbital phase and inclination. Figure 8 shows the ceiling as a selection function of planet period and host mass; the host-mass dependence is weak ($a \propto M_*^{1/3}$ at fixed period), so it is effectively a function of period alone. Tables 5, 12 and 10 give the detectability axes, the per-band decomposition and the selection function. Coverage eligibility and search completeness differ: the public usable ALMA archive (§3) sets eligibility, and every held execution block in scope was searched, so no EB-level selection lies behind the 102. A proper-motion-matched archive audit (2026 September 9) measures availability directly, counting 107 unique public on-star execution blocks and re-finding all but one searched block.

The survey sits against Wright et al. (2018)’s haystack axis by axis. On frequency (10 MHz–115 GHz) only Band 3 falls inside, 20.6 of 115 GHz, 0.18 of the axis, where Hz-resolution surveys reach far more finely; the remaining 66.6 GHz (Bands 4–8, 125–495 GHz) lies wholly outside and is the new territory. On transmitted bandwidth (0–20 MHz) a native-channel search probes a fraction set by channel width (window-mean 0.63, median 0.78). On sensitivity the median threshold (6.7 mJy across a 15.625-MHz channel, $\phi \approx 1.0 \times 10^{-21} \text{ W m}^{-2}$) detects a 10^{13} W transmitter only within $\sim 0.9 \text{ pc}$. Against the sole prior ALMA-archival search (Mason et al. 2024, 28 bycatch targets at $\geq 1.01 \text{ kpc}$, deepest $\text{EIRP}_{\min} 6.91 \times 10^{17} \text{ W}$), the 81 systems here, 41 inside 20 pc, reach median per-target thresholds $\sim 1250\times$ deeper. These fractions are not combined into one coverage: the axes are strongly coupled, distance and sensitivity above all, so a product of marginals would misstate the overlap by orders of magnitude.

What the zero detection bounds is conditioned on each target’s searched frequencies, the epochs observed and the searched morphology class. Turning it into a fraction of systems requires a per-system detection probability, which is not a demographic occurrence rate. We call it the *conditional searched-domain transmitter fraction* throughout, never “occurrence” or “transmitter fraction” unqualified, and keep its evaluation out of the interpretive argument: the framework, the demonstration and its sensitivity analysis sit in Appendix K, labelled *illustrative calculation only* and among no headline quantity (Table 5). The result of this paper is the search sensitivity, the exposure and the null. The demotion is measured, not rhetorical: completeness for the drift-resolved class rests on one injected configuration, and the error of transferring it to the other 118 Class A windows is unbounded, so a factor-of-two systematic would rescale every fraction derived from it.

In plain terms: as a by-product of other science, ALMA has stared at 88 of the nearest stars across parts of the millimetre and submillimetre band that no technosignature search had examined. We looked for a narrow, artificial-looking spectral line at each star, and found none the surrounding sky could not also produce. The powers we could have seen are enormous by human standards, and we could look only in the few gigahertz and the few hours each star happened to be observed. A non-detection on those terms says little about how common transmitters are, and a good deal about how to search an archive for them.

Because selection is on archival coverage alone, useful subsamples fall out without compromising the archive-complete framing, subject to the unresolved-pair caveat of §3. 18 of the 88 processed stars (41 known planets) are confirmed exoplanet hosts, among them Proxima Centauri, τ Ceti, ϵ Eridani, β Pictoris and the seven-planet TRAPPIST-1 system; HN Lib and LHS 1140, each with a habitable-zone planet, now carry an ALMA technosignature non-detection. Four searched stars are white dwarfs (Gentile Fusillo et al. 2021): Sirius B, Van Maanen’s Star, Wolf 219 and CD–38 10980, all continuum non-detections with nominal $\text{EIRP}_{5\sigma}$ trigger thresholds of 2.3×10^{13} – $8.4 \times 10^{14} \text{ W}$, among the few mm/submm SETI constraints for white dwarfs (Huang et al. 2026).

6.3. ALMA as a technosignature instrument: strengths and limits

ALMA offers four advantages. Aperture and receiver temperature give mm-wave sensitivities no single dish reaches. The tuning range, 125–495 GHz here, is essentially unsearched. Interferometric spatial discrimination, the control statistic of §4.1, tests a feature’s position against 512 alternatives, which single-dish and beamformed searches cannot. And the archive supplies 102 public execution blocks re-processable without new allocations. Each strength has its mirror. Native channelisation reaches 15.3 kHz at best, orders coarser than the Hz resolution of dedicated searches, so the search tests unresolved spectral excess, not fine structure; coverage is fragmented into 33 frequency islands weighted by other science; the line mask excludes 5.3 per cent of the union; configurations are heterogeneous, so no completeness transfers between them; and targeting is interest-driven rather than uniform (§3).

The largest unused advantage is the visibilities themselves. ALMA can test whether a spectral feature carries the interferometric phase signature of a celestial point source at a known position. For each surviving candidate one would compare competing likelihoods: $H_\star$, a source at the propagated stellar position, $V_{ij} = A(t, \nu) e^{-2\pi i(u_{ij}l_\star + v_{ij}m_\star)}$; H_{sky} , a source elsewhere; H_{common} , correlated or common-mode interference; and H_{noise} , thermal and calibration residual, and evaluate $\Lambda = \mathcal{L}(V|H_\star)/\mathcal{L}(V|H_{\text{noise}} + H_{\text{common}})$. This uses information single-dish and beamformed searches lack, is strictly more powerful than ranking a peak against a control ensemble, and would make that ensemble a calibration layer, not the primary authentication statistic. The search here is its amplitude-only prologue.

6.4. A next-generation ALMA technosignature search

Four changes would enlarge what this experiment can say, ranked by the signal space they buy.

Drift discrimination on coarse windows. Within the searched ceiling every in-grid drift on a 15.625-MHz window is sub-channel over a track, so 313 of 431 windows measure amplitude but cannot distinguish a drifting carrier from a stationary one (§4.4). Finer channelisation on re-observation, or a drift model spanning multiple execution blocks, would recover that axis.

Retain more control spectra. The two-stage scheme of §5.3, 512 controls routinely and a local null for any window that passes, fails wherever that null cannot be validated, because full dynamic spectra survive for only 8 control positions. Retaining all of them, at modest storage cost, would make the second stage usable in structured fields too.

Extend the injection campaign and the polarisation axis. The completeness curve should be measured per band, channel width, integration time, antenna count and primary-beam position rather than transferred by construction, and pushed past 10σ until it asymptotes, with EIRP₉₀ unlocated. The design is a stratified grid over those axes, injections at randomised drift rates and channel phases rather than preferentially on grid points, morphologies extended to sinusoidal acceleration tracks from representative sample orbits, trial-level logging from the first run, and an independent re-implementation cross-check of the injection code. The polarisation axis buys discrimination against weakly polarised astrophysical backgrounds at no sensitivity cost.

A boundary on drift linearity. The search assumes drift linear over a track. Curvature matters when $\frac{1}{2}\ddot{\nu}T^2$ exceeds half a native channel, reached only by the shortest-period orbits: $P \lesssim 1.6$ d for a typical Class A window (230 GHz, 488 kHz, $T \approx 2000$ s), about an hour for Class B. TRAPPIST-1 b, at 1.51 d, sits just inside the boundary and is also the case exceeding the drift ceiling; elsewhere the constant-drift model is adequate, and the chirp-search extension (Appendix C) covers the remainder.

7. CONCLUSIONS

This ALMA-archival technosignature survey searched, within 40 pc, every star with public usable coverage: 107 target/bands, 88 stars in 81 independent systems, ALMA-archive-complete and audited for position and identity (§5.2), but covering ~ 1 per cent of the catalogued stars in that volume and inheriting the archive’s non-uniform targeting. The archive’s channelisation splits the search: 118 Class A windows cover drift-resolved spectral carriers, 313 Class B windows only unresolved spectral excess, leaving no drift discrimination over 73 per cent of the survey (§4.1). Four windows raised a stage-1 statistical screening flag. Three are consistent with, and very likely caused by, CO emission: two circumstellar components towards β Pictoris across two transitions and three epochs, and one previously reported foreground cloud towards HD 48370 (Cataldi et al. 2023). The fourth, towards CP–72 2713 at 344.269746 GHz, is unclassified and marginal: a second-stage local null on its own dynamic spectrum gives $p = 1.6 \times 10^{-3}$, about 13 times above the survey-wide Bonferroni scale, so it is not significant at the survey level, and it is held under pinned promote and retire criteria (§5.3). None survives vetting, so we report 0 credible technosignatures. Barnard’s Star and Wolf 359 carry their first published mm/submm technosignature limits (§5.1), and forty ancillary continuum measurements (median 5σ limit 0.52 mJy) are non-detections of unquantified statistical power.

The EIRP_{5 σ} values are nominal trigger thresholds (boxed rule, §4): 2.3×10^{13} – 6.8×10^{16} W over Class A at 15.3–1953 kHz native resolution and 1.6×10^{13} – 2.8×10^{17} W over Class B at 15.625–31.25 MHz, deep enough toward the nearest system for an unresolved carrier matching the total power of an Arecibo-like planetary radar, and optimistic by $\times 2.00$ to $\times 2.7$ under the Hanning response. The conditional searched-domain transmitter fraction stays illustrative (Appendix K). Searched is not the same as constrained: the *Scope of the constraints* box of §4 fixes what the non-detection binds; outside it nothing is bounded. The first-priority extensions (§6.4) are drift discrimination on coarse windows, a visibility-domain likelihood ratio, control ensembles past the 1/513 rank floor, and a stratified injection campaign over band, channel width, integration time and beam offset. By-products ship with the data: a serendipitous molecular-line catalogue, exoplanet-host and white-dwarf subsample limits, and a closure-phase diagnostic, demonstrated but without discriminating power at these flux densities.

ACKNOWLEDGEMENTS

This paper uses ALMA data from the public ALMA Science Archive. The 102 execution blocks searched, with project codes and epochs, are listed in the machine-readable release and should be cited from it in the standard JAO.ALMA#XXXX.X.NNNNN.S form. They span 2013 October 6 to 2025 June 11 across 36 proposal codes. The AU Mic crossing window of §5.4 comes from the 2014 August 18 block of project 2012.1.00198.S, and one further public block of that project (uid://A002/X7d80c7/X15c6, 2014 June 5) also covers the 230.83 GHz window and is the second epoch of its recurrence test. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), MOST and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia*, processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database, operated at CDS, Strasbourg, France.

COMPETING INTERESTS

G.J.W. declares no competing interests. R.D. is an employee of VBRL Holdings Inc. (a private software company with no astronomy or SETI-related product line), which provided no funding for and took no role in this work. The authors declare no competing interests.

AUTHOR CONTRIBUTIONS (CREDIT)

Glenn J. White: Conceptualisation, Methodology, Supervision, Writing – review & editing.

Robin Dey: Software, Investigation, Formal analysis, Data curation, Visualisation, Writing – original draft.

DATA AVAILABILITY

The reduced data products underlying this paper, with the full target-selection and analysis pipeline, are at <https://github.com/Tilanthi/SETI>. All numerical values here come from the *submitted-analysis snapshot*: the repository commit tagged at submission freezes pipeline code and input data as run, is retained unchanged, and reproduces every number, table and figure from the regeneration scripts shipped with it.

The machine-readable catalogue, `per_target_results_v3.42.csv`, carries 431 rows, one per searched window, in 41 columns. Each row gives star name and machine-readable `system_id` (the independent stellar system that is the statistical unit, seven bound pairs sharing one identifier), distance, ALMA band and execution block, searched frequency range, native channel width, on-source time and integration count, per-channel rms, the nominal EIRP_{5 σ} threshold with both Hanning-corrected values (§4), the drift ceiling as a rate and as an acceleration, the trial drift count, η_{drift} , η_{smear} , resolution and search class, stellar and control peak statistics with the count of controls at or above the star and the add-one rank probability, the crossing frequency, the nearest catalogued transition with its offset in MHz and km s^{-1} , the full control geometry (ring centre, θ_{PB} , r_{in} , r_{out} , control count, seed) and the disposition, verbatim from Table 6. Three defects disclosed in earlier versions are repaired rather than documented: HD 139084B is no longer counted as two systems, so 81 is the independent-system total and every per-system denominator follows it; the empty `n_ant` column is removed, per-window antenna counts being recoverable only from the raw measurement sets, not from any retained product; and the disposition strings now match Table 6.

The complete release will also be deposited on Zenodo, with a DOI minted on final acceptance and inserted here. It comprises (i) per-window search products for all 431 windows: spectral extracts, per-drift-trial peak statistics, the machine-readable line-exclusion mask in both its frozen and repaired forms (§5.3), archive UID, epoch-propagated target coordinates and field centre, primary-beam gain, channel frequencies and widths, the drift grid, and the target statistic with all 512 control statistics from which the rank and p -value are recomputable; (ii) control-ensemble calibration products at all 512 positions; (iii) closure-phase and population-anomaly outputs; (iv) the full injection-recovery campaign products, all 1200 trials of Appendix B and all 3888 dwell trials, together with the second-stage local-null code, its per-window outputs and the retained null-draw arrays (§5.3); (v) the frozen pipeline code and input data snapshot with its change-log; (vi) the execution-block metadata and integration-time audit tables; (vii) the serendipitous molecular-line catalogue; and (viii) the proper-motion-matched execution-block availability census with its per-star ledger. Data products carry the Creative Commons Attribution 4.0 International licence (CC BY 4.0), and the pipeline code the MIT licence; neither author’s employer asserts any intellectual-property claim over the released code. Raw ALMA visibility data are public from the ALMA Science Archive at <https://almascience.org>.

SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; calibration, imaging and the local reprocessing of §§5.3–5.4), Python 3 with astropy, numpy and scipy (spectral extraction and statistics), and the ALMA Science Archive and SIMBAD TAP interfaces; all are free and open-source, with exact versions recorded in the frozen release.

APPENDIX

A. FREQUENCY CONVENTIONS, LINEWIDTH READING, AND SMEARING

Reference frames. The line mask’s chain runs topocentric $\rightarrow$ barycentric (Earth ephemeris at the execution-block mid-time) $\rightarrow$ stellar rest frame (catalogued systemic radial velocity), with JPL/CDMS rest frequencies (Pickett et

al. 1998; Müller et al. 2005); in the LSR frame it stops at the standard solar motion, not the systemic velocity (§5.3). The residual error is the catalogue radial-velocity uncertainty alone, a few km s^{-1} within 40 pc; second-order relativistic terms at the 50 km s^{-1} scale are $\lesssim 5 \text{ kHz}$ at 350 GHz, below the finest native channel searched (15.3 kHz). The β Pictoris and AU Mic attributions use the same chain, so their offsets are comparable to the exclusion tolerance.

Channel widths, and the instrumental response. These come from the measurement sets’ own spectral window tables (CHAN_WIDTH). Earlier versions read a missing “EFFECTIVE_BANDWIDTH” column as evidence of no online spectral averaging, hence that CHAN_WIDTH is also the effective resolution. MS v2 defines no such column; the required EFFECTIVE_BW and RESOLUTION are always present. The check is withdrawn, and with it the claim that the quoted channelisations are noise bandwidths.

ALMA applies online Hanning smoothing by default, so the spectral response is ~ 2 channels wide and adjacent-channel noise is correlated at $\rho \simeq 0.5$. Under the 0.25/0.5/0.25 kernel a sub-channel tone keeps 0.50 of its power in the peak channel at the channel centre and 0.38 on a boundary, median 0.44 over uniform placement. Every $\text{EIRP}_{5\sigma}$ here is therefore optimistic by $\times 2.00$ to $\times 2.7$, median $\times 2.29$; the released catalogue carries the nominal threshold and both corrected values per window, and the sensitivity summary quotes the worst case. The injections deposit an unsmoothed delta-in-channel tone, so the recovery curves of §4.4 and Appendix B carry the same bias and the end-to-end check (Eq. 4) cannot see it. Part of the lag-one correlation attributed to the drift-tracked statistic is instrumental on the same grounds, a decomposition not made here. We do not rescale: the per-window smoothing state sits in archive metadata not yet harvested, so each threshold here is nominal, and up to $\times 2.7$ optimistic if the window is Hanning-smoothed.

Linewidth convention. The statistic is the single-channel flux density after drift correction, so the quoted $\text{EIRP}_{5\sigma}$ applies to any transmitter narrower than the native channel, flat in the assumed linewidth, with two sub-channel qualifications. Flatness is exact only for a centred tone, a boundary-straddling one splitting its power between two channels at worst by a factor of two; and the channel-to-channel transition follows the correlator’s passband shape, not retained in the frozen products, so flatness is a channel-centred idealisation good to tens of per cent. Thresholds therefore carry a sub-channel-position systematic of that size, a randomly placed tone keeping a mean 0.75 of its power in the stronger channel, and EIRP values are quoted to two significant figures.

Transmitters wider than one channel. The baseline filter is a running median along frequency, flat for features narrower than roughly half the median width W and falling to zero above W , so the survey’s effective sensitivity is to channel-confined morphologies and W is the linewidth ceiling. W is not carried in the frozen per-window products, so it cannot be printed here; it ships with the tagged release. Within that ceiling, a top-hat transmitter spanning N_{ch} native channels has per-channel flux the total divided by N_{ch} at unchanged noise, so to first order

$$\text{EIRP}_{5\sigma}(N_{\text{ch}}) \simeq N_{\text{ch}} \times \text{EIRP}_{5\sigma}(1), \quad N_{\text{ch}} \lesssim W/2, \quad (\text{A1})$$

the rescaling the main text applies to signals broader than one channel.

Worked example: from flux to EIRP. Take UV Ceti (G 272-61B) Band 3. The searched window’s per-channel rms is 0.244 mJy, so $S_{\text{min}} = 5 \times 0.244 = 1.22 \text{ mJy}$ in one 15.625-MHz native channel, or $1.22 \times 10^{-29} \text{ W m}^{-2} \text{ Hz}^{-1}$ at 2.67 pc, and $\text{EIRP}_{5\sigma} = 4\pi d^2 S_{\text{min}} \Delta\nu = 1.6 \times 10^{13} \text{ W}$.

Intra-integration smearing. The frequency excursion at the maximum searched drift rate across one integration, $\dot{\nu}_{\text{max}} t_{\text{int}}$, is at most 1.10 channels across the 431 windows (median 0.001, 90th percentile 0.06). A linear drift smeared by Δ channels within an integration keeps a fraction $\eta_{\text{smear}} = \text{sinc}(\pi\Delta/2)$ of the peak-channel amplitude, so the effective threshold is $\text{EIRP}_{5\sigma}/\eta_{\text{smear}}$. Three windows have $\eta_{\text{smear}} < 0.99$: β Pic Band 6 ($\Delta = 1.10$, $\eta_{\text{smear}} = 0.574$), η Crv Band 8 (0.585, 0.865) and η Crv Band 7 (0.411, 0.932). Every other window agrees with its nominal threshold to better than 1 per cent. A signal drifting midway between adjacent trial rates accumulates at most 0.03 channels of residual drift across the grid as realised, so no significant sensitivity is lost between them.

B. INJECTION-RECOVERY SENSITIVITY VALIDATION

To test whether a 5σ threshold behaves as idealised Gaussian noise predicts, we inject synthetic tones of known amplitude into real visibilities, through the extraction code’s own mechanism and as far upstream of the statistic as retained products allow, and measure what the unmodified pipeline returns. Three results inherit its single-configuration limitation: the drifting-class recovery of Appendix K, the machinery and boundary-placement amplitude factors of §4.4, and the coarse-window dwell reading of §4.4.

Six configurations span three bands and both classes: three coarse Band 7 spws of AT Mic, one coarse spw each in Bands 3 and 6 of AU Mic, and AU Mic’s 488-kHz flagband window with its 49-trial drift grid. Each receives a complex tone at the star position, at amplitudes 4–10 $\times$ the statistic’s own noise, drift fractions $f_{\text{drift}} = 0, 0.25, 0.5, 0.75, 1.0$ of the ceiling, on- and half-grid rates, and channel-centred and boundary placements: 200 trials per configuration, 1200 in all, under a frozen criterion ($T \geq 5$, peak channel within 3 of the injection, peak drift within one grid step).

The coarse-window variant recovered 0 of 500 injections at every amplitude from 4σ to 10σ under that criterion. This is a recovery-criterion artefact, not a pipeline suppression, and it is the one withdrawn result here: the drift-matching clause cannot be satisfied on coarse windows, where every in-grid drift is sub-channel over the track, so the reported peak drift is arbitrary. Scored on detection alone the same windows recover essentially everything, and an end-to-end injection through the released pipeline returns T_* of 20.5, 65.3 and 217.7 at 1, 3 and 10 times the per-integration noise, against an ideal coherent gain of $\sqrt{639} = 25$. Class B limits therefore bound carriers of any dwell fraction, with the completeness of §4.4; what those windows lack is drift discrimination. The fine window fails the same criterion the

TABLE 10

SELECTION FUNCTION OF THE SEARCHED SAMPLE AGAINST A VOLUME-LIMITED REFERENCE CENSUS. REFERENCE CENSUS: GAIA DR3 SOURCES WITH $\varpi > 0$, $\sigma_\varpi/\varpi \leq 0.1$ AND $1/\varpi \leq 40$ pc (17566 OBJECTS; 8594, 49 PER CENT, CARRY A **TEFF_GSPPHOT** ESTIMATE). SAMPLE: THE 88 STARS WITH AT LEAST ONE SEARCHED WINDOW, AT THE DISTANCES THE SEARCH ITSELF USED. TEMPERATURES ARE AVAILABLE FOR 52 OF THE 88 STARS (ALL 88 MATCHED TO THE 168-ENTRY ALMA-COVERED CENSUS, 58 BY EXACT NAME; 52 OF THOSE MATCHES CARRY A TEMPERATURE), SO THE TEMPERATURE PANEL IS NORMALISED WITHIN EACH COLUMN OVER CLASSIFIED OBJECTS ONLY, WITH T_{eff} A PROXY FOR SPECTRAL CLASS. “RATIO” IS THE SAMPLE COLUMN FRACTION OVER THE CENSUS COLUMN FRACTION: 1 MEANS THE SAMPLE TRACKS THE REFERENCE POPULATION, > 1 OVER-REPRESENTATION.

Property	Census	Searched	Ratio
Distance (all 88 searched stars; 17566 census objects)			
$d \leq 5$ pc	48 (0.3%)	12 (13.6%)	49.9
5–10 pc	240 (1.4%)	13 (14.8%)	10.8
10–20 pc	2069 (11.8%)	18 (20.5%)	1.7
20–30 pc	5391 (30.7%)	18 (20.5%)	0.7
30–40 pc	9818 (55.9%)	27 (30.7%)	0.5
T_{eff} class ^a (52 searched; 8594 census)			
A (≥ 7500 K)	24 (0.3%)	1 (1.9%)	6.9
F (6000–7500 K)	402 (4.7%)	9 (17.3%)	3.7
G (5300–6000 K)	860 (10.0%)	16 (30.8%)	3.1
K (3900–5300 K)	1400 (16.3%)	5 (9.6%)	0.6
M (< 3900 K)	5908 (68.7%)	21 (40.4%)	0.6
Other axes			
Known planet hosts ^b	20 (11.9%)	18 (20.5%)	1.7
Searched stars / systems	–	88 / 81	–

^aBins are temperature bins, not MK types: A ≥ 7500 , F 6000–7500, G 5300–6000, K 3900–5300, M < 3900 K. Gaia **teff_gspphot** is unavailable for about half the reference census, preferentially for the coolest and faintest objects, so the census M fraction is a lower limit and the earlier-type ratios are correspondingly upper limits. White dwarfs are not separable in this parameterisation and fall wherever their (unreliable) photometric temperature places them.

^bThe Gaia reference census carries no planet information; the census column here is instead the 168-entry ALMA-covered census, of which 20 entries are known planet hosts. The 88 searched stars include 18 hosts of 41 known planets.

TABLE 11

AMPLITUDE COMPLETENESS AT THE TRIGGER, BY WINDOW CLASS AND CARRIER MORPHOLOGY, FROM THE 1200 INJECTIONS OF THIS APPENDIX. “DETECTION” IS AMPLITUDE RECOVERY AT THE NOMINAL TRIGGER; “DRIFT” IS WHETHER A RECOVERED DRIFT RATE IS INFORMATIVE ABOUT THE TRANSMITTER; N COUNTS THE SEARCHED WINDOWS OF THAT CLASS. THE SECOND EXPERIMENT, TABLE 4, IS *not* A THRESHOLD COMPLETENESS: IT INJECTS NON-DRIFTING CARRIERS AT $\geq 4\sigma$ PER INTEGRATION INTO 12 REAL WINDOWS AND MEASURES TEMPORAL OCCUPANCY. THE TWO MUST NOT BE READ ACROSS.

Class	Morphology	Detection	Drift	N
Class A (fine)	drifting	measured: $C_{5\sigma} = 0.42$	measured	118
Class A (fine)	near-static	post-baseline only ^a	n/a ^b	118
Class B (coarse)	any	end-to-end	none	313

^aThe dwell campaign recovers 91–100 per cent at $f_{\text{dwell}} \geq 0.25$, injecting after the baseline step; through-pipeline recovery is unmeasured.

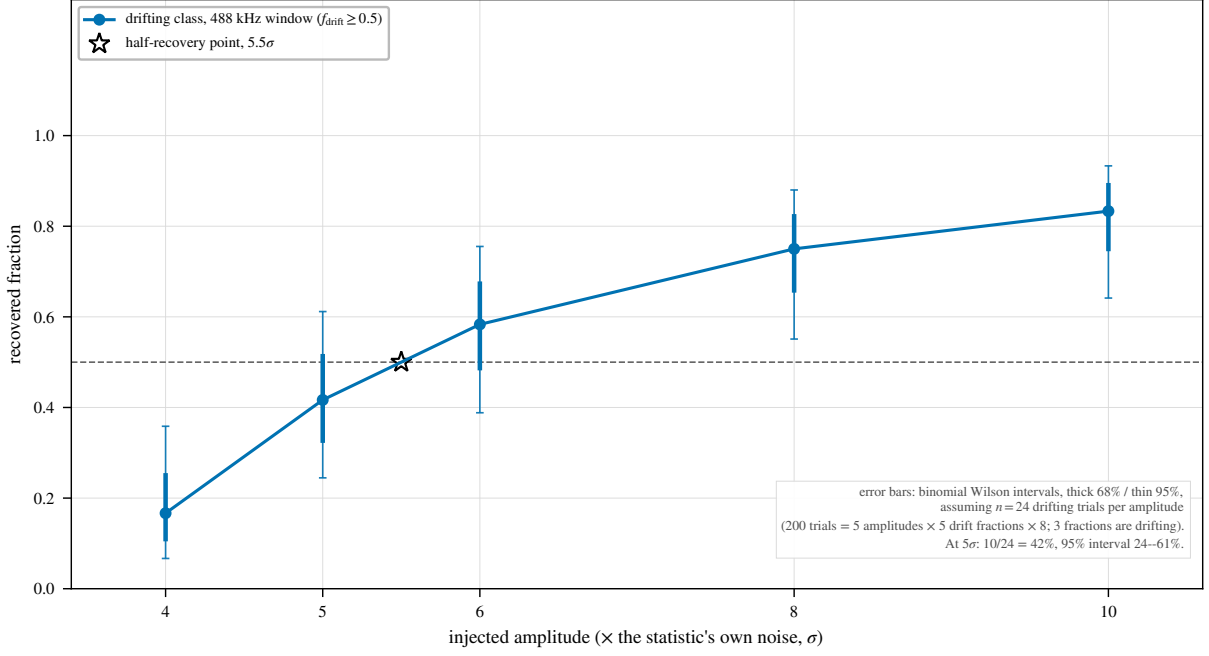
^bSub-channel drift over the track by definition.

same way, and that cell is unmeasured rather than lost: carriers with $f_{\text{drift}} \leq 0.25$ dwell ≥ 40 per cent of the track per channel and fail it, those with $f_{\text{drift}} \geq 0.5$ surviving as the searched class.

The second result is the completeness curve. On the fine window, drifting carriers ($f_{\text{drift}} \geq 0.5$, pooled over grid phase and sub-channel placement) are recovered 17, 42, 58, 75 and 83 per cent at 4, 5, 6, 8 and 10σ , saturating by twice threshold power (Fig. 7). The machinery alone, with the tone injected after the baseline step, is amplitude-faithful to within $-9/+14$ per cent for channel-centred tones at every drift fraction. Boundary-straddling placement costs a factor 0.6–0.8 in the statistic and recovers 0 per cent at 5σ against 83 per cent for centred ones (67 and 100 per cent at 10σ), crossing 50 per cent near $1.4\times$ threshold amplitude. Survey completeness uses the placement-pooled curve, conservative for the boundary-straddling case: $\text{EIRP}_{50} = 1.10 \text{EIRP}_{5\sigma}$ and $\text{EIRP}_{90} > 2.0 \text{EIRP}_{5\sigma}$, a bound only, both carrying an unbounded transfer systematic to every configuration but this one, bracketed as $0.5\text{--}2\times$ in Appendix K. Not spanned: other target environments, primary-beam offsets, integration durations and edge-phased drift grids; bounding that transfer error, propagated in Appendix K, needs the multi-configuration campaign of §6.4.

C. DEMONSTRATED DIAGNOSTICS WITH NO OPERATIVE ROLE HERE

Closure phase. The closure phase $\arg(V_{ij}) + \arg(V_{jk}) - \arg(V_{ik})$ vanishes for an unresolved point source anywhere on sky and is immune to per-station calibration errors, so it tests only point-source coherence; a coherent far-sidelobe interferer would pass. We score consistency with zero by a circular (unit-phaser) z-score of the mean drift-tracked closure phase, nulled on each observation’s own integrations, validated on synthetic visibilities and run on continuum detections in four local reprocessing fields, including the resolved AT Mic Band 7 binary as a resolving control. At



Measured injection recovery. Denominators are reconstructed from the campaign design, not recorded in the retained products: the fine-window value $n=24$ is the unique design-consistent choice that reproduces all five quoted percentages exactly.

FIG. 7.— Measured injection recovery on real ALMA data, fine-channel configuration. Recovery $C(A, f_{\text{drift}})$ rises with amplitude for drifting carriers; static and near-static ones fail the frozen criterion, and boundary-straddling tones can fail its peak-drift clause while still being detected. Coarse configurations are not plotted: under the same criterion they recover nothing, withdrawn in the text as an artefact of the criterion, not a property of the pipeline.

these flux densities it is useless: per-baseline S/N is 0.5–0.7 at the median and never exceeds 5.1, only 0.02 per cent of triangles have all three baselines above S/N=5, and every field’s closure phases, the binary included, are indistinguishable from noise. It discriminates only at greater depth.

Chirped drift and periodicity. Stacking at constant linear drift rate is blind by design to a drift rate that changes during the observation, and to pulsed or periodically modulated carriers. We implemented a chirped (quadratic-drift) extension on a coarser grid of 21 linear-drift $\times$ 5 chirp-rate trials, and a per-channel periodicity search, a per-position FFT along each channel’s time axis. Both vet the star against 8 retained control positions, a strict subset of that window’s 512-position ensemble, not an independent one. On the 122 retained spectra this is a demonstration: flag rates of 10.7 per cent for chirp and 13.1 per cent for periodicity against a ~ 11 per cent chance rate show only that the extension runs correctly under the null. No instance is treated as a candidate and no limit derives from it; extending both to the full ensemble would make the screen survey grade.

D. CROSS-TARGET FREQUENCY-OCCUPANCY CHECK

With no ON/OFF cadence against radio-frequency interference, this single-pointing archival survey substitutes the many unrelated stars sharing similar correlator setups: a channel-confined feature recurring at the same observed *sky frequency* towards different targets is almost certainly interference or instrumental. Over every target/spw result the check pairs the best-fit-channel frequency and the formal $\geq 5\sigma$ crossings, matching ν_1, ν_2 towards different targets when $|\nu_1 - \nu_2| < \max(k \Delta\nu_1, k \Delta\nu_2)$ with $k = 3$ (~ 46 kHz). Across the 431 windows and 107 tabulated crossings, no pair falls within the kernel, or within 0.5 MHz. The kernel is narrower than the β Pictoris line offsets of 10.6 and 21.9 MHz, so those windows were dispositioned by line-catalogue inspection.

The null’s power is the count of target pairs sharing frequency coverage: on the ≤ 20 pc subset, 366 of 946 star pairs (39 per cent) overlap by at least 0.01 GHz, deepest 14.25 GHz (γ Lep– γ Vir), and 109 pairs involve one of the five stars carrying formal crossings, so every crossing frequency was tested against overlapping independent targets. The UV Ceti and AT Mic shared execution blocks are excluded, their intra-pair coincidences being identities by construction. The interim product’s per-window crossing lists are capped at 50 frequencies, so the true count is 171 crossings against 107 tabulated. The difference is one window, β Pic Band 6 ultra-fine, whose 114 formal crossings enter as the 50 strongest; its untabulated 64 could not be tested, so the null is demonstrated for the tabulated set and unresolved for the rest. The public release ships uncapped crossing lists.

E. PER-TARGET RESULTS

A typeset table would run to many pages, so these results ship with the manuscript source as `per_target_results_v3.42.csv`, one row per searched window, summarised in the Data Availability statement. The $\text{EIRP}_{5\sigma}$ column uses the definition of §4, on the same threshold and primary-beam correction as the continuum column, so a row’s two limits are comparable; the per-channel flux-density threshold follows from it, using the carrier lane’s per-channel noise,

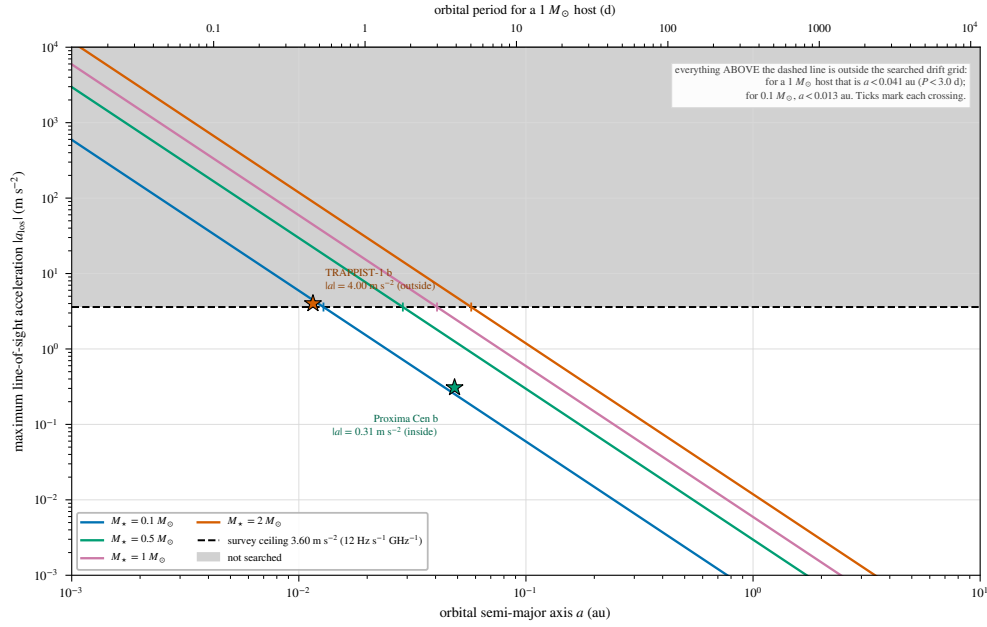


FIG. 8.— The drift-rate ceiling as a physical selection function. Maximum star–planet relative acceleration, edge-on and maximised over orbital phase, as the carrier drift rate it induces (Hz s^{-1} per GHz of carrier frequency; equivalently $\dot{\nu}/\nu = \dot{v}/c$), against orbital period, for host-star masses $0.09\text{--}1.8 M_{\odot}$ (lines) and the known planets of the planet-hosting targets (points). Shaded band: the survey’s drift-search ceiling, above which a transmitter lies outside the searched grid. Only TRAPPIST-1 b reaches it (§6).

TABLE 12

PER-BAND SURVEY CHARACTERISATION OF THE 431 SEARCHED WINDOWS. “SYS.” COUNTS UNIQUE STELLAR SYSTEMS SEARCHED IN THE BAND, SO A SYSTEM OBSERVED IN TWO BANDS IS COUNTED IN BOTH ROWS (81 SYSTEMS, 88 STARS IN TOTAL); “EBs” THE CONTRIBUTING EXECUTION BLOCKS, ATTRIBUTED BY THE PROVENANCE OF THE DE-DUPLICATED WINDOWS; $\Delta\nu_{\text{ch}}$ THE NATIVE CHANNEL WIDTH IN MHz (MEDIAN, RANGE IN BRACKETS); “UNION” THE UNIQUE SKY FREQUENCY SEARCHED IN THE BAND AFTER MERGING OVERLAPPING WINDOWS; $\text{EIRP}_{5\sigma}$ THE MEDIAN NOMINAL THRESHOLD ACROSS THE BAND’S WINDOWS; $\dot{\nu}_{\text{ceil}}$ THE RANGE OF DRIFT-RATE CEILINGS IN Hz s^{-1} PER GHz OF CARRIER FREQUENCY. THE “ALL” ROW IS COMPUTED OVER THE WHOLE SAMPLE, NOT SUMMED DOWN THE COLUMNS: EBs AND SYSTEMS RECUR ACROSS BANDS AND THE FREQUENCY UNIONS ARE DISJOINT BY CONSTRUCTION.

Band	Sys.	EBs	Win.	$\Delta\nu_{\text{ch}}$ (MHz)	Union (GHz)	Med. $\text{EIRP}_{5\sigma}$ (W)	$\dot{\nu}_{\text{ceil}}$
3	8	9	40	15.63 (0.244–15.63)	20.6	1.1×10^{14}	12.0–13.3
4	1	1	4	15.63	6.9	2.6×10^{13}	12.0
5	1	1	4	15.63	6.9	4.4×10^{13}	12.0
6	54	54	216	15.63 (0.015–31.25)	29.4	1.3×10^{15}	12.0–13.3
7	33	34	155	15.63 (0.061–15.63)	22.6	2.9×10^{15}	12.0–13.3
8	3	3	12	15.63 (0.061–15.63)	6.6	6.1×10^{16}	12.0
All	81	102	431	15.63 (0.015–31.25)	93.1	1.4×10^{15}	12.0–13.3

not the tabulated rms, which is that target’s line-excluded full-bandwidth continuum noise. Bands 6 and 7 supply 86 per cent of the windows, an archive imprint, not a design choice.

F. THE VELOCITY-SPACE LINE MASK: JUSTIFICATION AND RETROSPECTIVE APPLICATION

The one-channel line check failed the β Pictoris windows (§5.3): a native channel spans a fraction of a km s^{-1} , while circumstellar kinematics shift real emission by many channels. The replacement mask takes its tolerance from the kinematic budget of circumstellar gas: Keplerian motion where these molecules survive ($\gtrsim 10$ au), $\lesssim 13 \text{ km s}^{-1}$; cold-gas linewidths $\lesssim 1 \text{ km s}^{-1}$; cold dense-gas tracers with no bulk-wind term for the dwarfs dominating the sample; and an uncorrected barycentric term bounded by $\pm 30 \text{ km s}^{-1}$. The result is empirically conservative: the largest attributed circumstellar component (β Pic, -2.7 to -3.6 km s^{-1}) sits at 7 per cent of it, and *zero* crossing frequencies lie 13–50 km s^{-1} from their nearest transition, so a $\pm 13 \text{ km s}^{-1}$ Keplerian-only mask would have suppressed and released the same windows: the wider tolerance costs no yield and buys robustness against radial-velocity catalogue error.

The repaired mask. The mask’s four defects as executed, and the retrospective re-application that repairs them without re-running the survey, are in §5.3; no disposition changes. The rounding defect is also why the released CO(1→0) offset column, referred to the frozen mask’s rounded 115.271000 GHz entry, reads 0.202 MHz (0.53 km s^{-1}) smaller than the laboratory-referred value of Table 8. A cheaper, better-founded rule than enumerating species: the archive records the rest frequency each execution block was tuned to, so masking each window’s own tuned line would catch the β Pictoris, HD 48370 and [C I] windows automatically, alongside an LSR-frame tube of a few $\times 10 \text{ km s}^{-1}$.

Two of the six remaining crossing windows fall inside the mask (the unflagged β Pic coarse window at -3.6 km s^{-1} ; HD 285968 at $+8.7 \text{ km s}^{-1}$, already rejected by its control ring, maximum 10.9 against a star peak of 6.5); the rest lie

far outside any circumstellar tolerance (AU Mic +379, LHS 1140 +115 km s⁻¹). Of the 20 windows with an on-star crossing, 4 also beat their control ring, three accounted for by the mask. The other 16 are ordinary noise crossings: in each, the window’s own control ring reaches or exceeds the stellar value (ring-maximum median 5.84 against a crossing median 5.18), the expected Gaussian-tailed population of moderate 5–5.5 σ crossings with comparably strong peaks among 512 controls. But the transitions that make astrophysical false positives cheap are also natural reference frequencies for a deliberate communicator (Cocconi & Morrison 1959), so the released products keep a known-line-coincident list for a future search with independent discrimination: every on-source 5 σ crossing inside the masked intervals, with its offset from each nearby transition.

G. AU MIC: THE THREE TESTS IN FULL

Localisation. Reprocessed from the raw science data model with the observatory pipeline’s own calibration recipe (310 integrations against the release pipeline’s 249, on a finer drift grid) and re-run through the symmetric statistic of §4.1, the crossing window’s star does not beat its control ensemble. The frozen products agree only asymmetrically: the source-region peak exceeds the 512-control maximum by 0.12, below that statistic’s window-to-window scatter, and the crossing fails the single-position variant too.

Recurrence. The one further public Band 6 block covering 230.83 GHz, the same programme executed 2014 June 5, was searched identically: no channel within ± 10 of the first-epoch flagged channel exceeds 3.1, and nothing is localised to the star. Its band rms (1.48 mJy per 488-kHz channel) matches the first epoch’s 1.9 mJy, so a comparable excess would have been recovered, and the frequency recurs at no other target. Intermittent or precisely-scheduled transmissions outside the sampled windows stay unconstrained.

Statistical scale. The on-source peak (5.97) sits at the 92.1th percentile of the control-maxima distribution over all 431 windows (median 4.56, 90th percentile 5.85): 34 windows (7.9 per cent) match or exceed it, add-one empirical $p = 0.081$ before any multiple-testing correction: a survey-wide window-level rate, distinct from the within-window rank of §5.4. The crossing matches no catalogued molecular transition (287 MHz from CO $J=2\rightarrow 1$), and closure-phase vetting finds it point-source consistent, one-sided evidence only (Appendix C).

Two astrophysical contexts are recorded for re-examination; neither grounds the rejection. AU Mic is an exceptionally active flare star, but the block’s own quiescent continuum (0.15 mJy) rules out the later-epoch GHz-wide flares of MacGregor et al. (2020) (16.8 and 6.1 mJy): an 11 mJy channel-confined excess would sit $\sim 70\times$ above a continuum that shows nothing. The controls share the primary-beam footprint, so smooth structure such as the debris belt raises star and controls alike and cannot alone produce a localised excess.

H. THE ANCILLARY CONTINUUM LANE

The continuum lane is an exploratory by-product, not a co-equal search: 57 measurements, 17 detections and 40 limits, a millimetre excess being at best an indirect, model-dependent observable of waste-heat technology. An “upper limit” is $5\times$ the primary-beam-corrected map rms at the star’s position, and the deepest non-detection is towards TRAPPIST-1 in Band 3. Under the combined error model sixteen of seventeen detections stay above 5 σ ; the exception (η Corvi Band 7, 4.7 σ combined) is marginal. Bright far-from-target peaks (ϵ Eri, GJ 674, CD–23 14742) fall outside the 3-arcsec tolerance and count as non-detections.

Every detection is astrophysically unsurprising: the unresolved UV Ceti binary; six systems whose millimetre flux is disc or chromospheric emission, among them the young, active β Pic, AU Mic and AT Mic, and the old, inactive τ Cet, a ~ 5 –8 Gyr G8V star whose flux is photospheric and/or its debris disc; six stars at photospheric flux, among them γ Lep, γ Vir, χ^1 Ori and η Cru; and two late-M dwarfs of documented magnetic activity, LP 349–25 and the auroral emitter LSR J1835+3259 (Hallinan et al. 2015). Closure phase cannot discriminate structure for any of the seventeen at these flux densities.

The anomaly screen. Each star’s flux is tested against its own photospheric expectation (Planck function with Gaia DR3 parameters, Pecaút & Mamajek fallback where GSP-Phot is unavailable) and against same- T_{eff} stars in coarse bins of at least five, by a robust median/MAD statistic. The error model, evaluated before any flag, is $\sigma_{\text{tot}}^2 = \sigma_{\text{map}}^2 + (f_{\text{cal}} S_{\nu})^2 + \sigma_{\text{model}}^2$, with f_{cal} 5 per cent in Bands 3–6 and 10 per cent in Bands 7–8 and σ_{model} propagated from Gaia temperature and radius errors and the fallback relation’s scatter. The model has *no intrinsic-variability term*: adequate for a photosphere, wrong for an active M dwarf whose millimetre flux can vary by factors of several, so the screen is uninformative for exactly the stars a radio-stellar reader cares about. At the five-star minimum the bin median rests on two or three values, so it flags for inspection rather than testing per-star hypotheses. One outlier results: χ^1 Ori at Band 3 (excess ratio 2.52, largest of the seventeen; resampled tail probability 0.53 in its six-star bin, so a ranked outlier of a small sample, not a significant departure), consistent with a spectroscopic G0V binary’s unresolved companion or a chromosphere.

I. FALSE-ALARM ACCOUNTING UNDER THE SYMMETRIC CRITERION

The per-window rate is 1/513, the survey expectation 431/513 = 0.84 flags. The floor falls only if the control ensemble grows, which the primary beam bounds; and windows are not fully independent, grouping into 102 execution blocks that share weather, calibration and correlator setup, so $N/513$ is an expectation over a mildly correlated sample, not N independent trials. Both limits are conservative for the conclusion drawn, that no window warrants candidate status: they widen rather than narrow the null range.

Over 4.60×10^7 channel $\times$ drift cells, a one-sided Gaussian 5σ tail (2.87×10^{-7}) predicts 13.2 chance crossings, against the 20 windows with an on-star crossing. Raw counts run higher because the drift-tracked statistic is strongly correlated between adjacent channels ($\rho_{\text{lag}1} = 0.989\text{--}0.991$, i.e. $8.5 \times 10^4\text{--}1.1 \times 10^5$ effectively independent cells), so crossings arrive in contiguous runs; collapsed to clusters on a kernel of 3 native channels they leave far fewer independent trials, in agreement with the trials expectation. Beyond the shared primary-beam gain, neighbouring control positions are correlated below the synthesised-beam scale, so independent spatial trials fall from 512 to $N_{\text{eff}} \simeq 30\text{--}43$, a floor to which the empirical 1.2 per cent rate, measured on the correlated ensembles themselves, is immune by construction.

No single stratum drives the star-exceeds-ring rate. The stratification was run on the ≤ 20 pc subset under the withdrawn asymmetric statistic, whose counts it quotes, the operative symmetric rate being too small (5 windows) to stratify. Across band, channelisation and primary-beam offset that rate moved only between 11 and 20 per cent, and the disc-hosting caveat inverts empirically: disc- or planet-hosting stars contribute every crossing window and every stage-1 flag but a *lower* star-exceeds-ring rate, 9.8 against 18.3 per cent for non-hosts, so circumstellar emission shared by star and ensemble does not drive the statistic. But that rate test is not the relevant one. Every flagged window, and the crossing the statistic revision removed, belongs to one population: β Pic, AU Mic and CP-72 2713 are β Pictoris moving-group members with debris discs and HD 48370 is a disc host behind a molecular cloud. The flags concentrate where the astrophysical foreground is richest, which is what the archive contains: the searched sample is, to a good approximation, the ALMA debris-disc and young-moving-group archive, the worst case for astrophysical false positives at millimetre wavelengths, and the flag statistics should be read accordingly. Excluding the 15 β Pictoris windows leaves one unattributed crossing among 416, against 0.81 expected ($P(\geq 1) = 56$ per cent).

J. EXCLUDED ENTRIES, AND THE RETIRED STATISTIC

Five pre-freeze changes are logged with their effect on released products in the repository change-log, the auditable record; the other four are the threshold convention, the withdrawn baseline-operator description, the line-exclusion width and the framing of the conditional searched-domain transmitter fraction, each treated where it belongs above. Only the replacement of the asymmetric region-max statistic by the symmetric single-position form of §4.1 (release re-run, criterion frozen) changes a released number: flagged windows 65/431 (15.1 per cent) become 5/431 (1.2 per cent), and the AU Mic crossing becomes unflagged. Under a pure-noise null the retired region maximum over n_{src} positions beats single controls with probability $n_{\text{src}}/(n_{\text{src}} + n_{\text{ctrl}}) = 20.9$ per cent, the order of the measured region-max rate, never interchanged with the symmetric one. Nothing derived from the retired statistic enters an operative false-alarm probability.

The four classes of excluded entry. (i) Four target/bands (Barnard’s Star [B7], Wolf 359 [B7], SCR J1845–6357, Wolf 358) have no carrier result from a process-level failure; each keeps a valid continuum non-detection and no EIRP threshold. (ii) Giclas 9–38A and Giclas 9–38B [B3] are excluded entirely, the pair’s only matched MOUS pointing ~ 52 arcmin from either star, $35\times$ its own primary-beam FWHM; the carrier step aborts correctly, continuum imaging lacks that guard. (iii) Six windows across five stars (51 Eri, HD 10647, HD 139664, TWA 7, HD 92945 twice) fail the noise test of §5.2: all are 15.62-MHz coarse windows whose reported rms is inconsistent with their reported integration by three to four orders of magnitude and sits a factor 388–669 below the shallowest sibling of the same block and channel width, a normalisation signature, not weather or configuration. Naming the root cause needs quantities carried only by the raw execution blocks, as does checking whether the 512 control σ values in a defective window are deflated with the stellar one. (iv) The catalogue entry “ γ Lupi” resolves to the F4V debris-disc host HD 139664 at 17.40 pc, not the naked-eye γ Lupi; all such entries are relabelled and no measured value changes. The supporting quality audit is clean across all 431 windows bar four χ^1 Ori Band 3 windows flagged as elevated ($0.9\text{--}1.1\times$ the Band 3 median, weakening their thresholds by $\lesssim 20$ per cent in amplitude).

K. THE CONDITIONAL SEARCHED-DOMAIN TRANSMITTER FRACTION

Illustrative calculation only. Nothing here is a survey result, and no number in it appears among the headline quantities of Table 5. It is kept for the framework, not for the number it returns.

A zero detection constrains a fraction of systems only through a per-system detection probability, factorised as

$$C_i(P) = F_i R_i(P) D_i C_{\text{morph},i}, \quad (\text{K1})$$

with F_i the chance the transmitter’s sky frequency lies in system i ’s searched intervals, $R_i(P)$ the chance a transmitter of EIRP P would have crossed that system’s threshold, D_i the chance it emitted during the analysed archival interval, and $C_{\text{morph},i}$ the measured recovery of a carrier of the searched drift and temporal morphology (Appendix B). The probability of zero credible detections if a fraction f of systems hosts one is

$$P(N_{\text{det}} = 0 | f) = \prod_{i=1}^{N_{\text{sys}}} (1 - f C_i), \quad (\text{K2})$$

and the one-sided 95 per cent upper limit f_{95} solves $P(N_{\text{det}} = 0 | f_{95}) = 0.05$ over the 81 systems.

Everything below sets $F_i = D_i = 1$ *by conditioning, not by measurement*, so f is a conditional searched-domain transmitter fraction and not a demographic occurrence rate. At EIRP $\geq 10^{16}$ W the measured form gives $f_{95} \approx 6$ per cent over the 58 systems carrying a Class A window, with idealised brackets 3.7 and 8.8 per cent. Folding in the dwell-measured Class B curve, so each window carries the completeness measured for the morphology it searches,

TABLE 13

CONDITIONAL SEARCHED-DOMAIN TRANSMITTER FRACTION, ILLUSTRATIVE CALCULATION ONLY. *These numbers assume every searched star was broadcasting continuously, in band, at the stated power during the observation. They are not a demographic limit on the 40 pc stellar population, nor on planet-hosting or Earth-like systems.* (A) THE 95 PER CENT UPPER LIMIT $f_{95}(P)$ ABOVE EIRP P , n THE SYSTEMS WITH NON-ZERO COMPLETENESS AT THAT P ; THE MEASURED COLUMN CARRIES THE INJECTION-MEASURED RECOVERY, WHOSE SINGLE-CONFIGURATION TRANSFER ERROR IS UNQUANTIFIED. (B) EPOCH-ACTIVITY DEPENDENCE AT $P \geq 10^{16}$ W, IN THE THRESHOLDED (81 SYSTEMS, UNIT RECOVERY) AND MEASURED-COMPLETENESS (58 CLASS A-CALIBRATED) FORMS. ‘UNCONSTRAINED’: NO BOUND EXISTS AT THAT COMBINATION.

(a)				
P (W)	thresholded (unit C)	measured C		
	n	f_{95}	n	f_{95}
3×10^{13}	4	52.7%	2	unconstrained
10^{14}	16	17.1%	11	47.1%
10^{15}	59	5.0%	47	9.5%
10^{16}	80	3.7%	58	6.3%
10^{17}	81	3.6%	59	6.0%

(b)			
Epoch activity	p_{epoch}	f_{95} thresholded	f_{95} measured
1 (continuous)		3.7%	6.3%
0.5		6.5%	11.0%
0.1		29.2%	48.8%
0.01		unconstrained	unconstrained

extends the calculation to 80 systems and moves the number to 4.3 per cent (9.4 per cent over 66 systems at 10^{15} W, 3.8 per cent over 81 at 10^{17} W). Table 13 carries the ladder.

Three properties bound how far this travels. The Class A recovery curve is measured on one configuration (AU Mic, 488-kHz Band 6) and applied to all 118 Class A windows, spanning Bands 3–8 and on-source times 21–5274 s; the transfer error is not bounded by the ${}^{+16}_{-14}$ percentage-point binomial interval, which is sampling noise within that one run. Scaling the curve by 0.5–2 $\times$ moves the number from 6.3 to 5.0–12.3 per cent, the honest width. The C_i are not independent: 431 windows group into 102 execution blocks with an intraclass correlation of 0.24 in the control-normalised stellar rank, and the design effect moves 6 to 7.1 per cent, an order below the transfer term. Under any transmitter-frequency prior spread across ALMA’s tuning range the bound vanishes: the per-system Class A union has median 1.73 GHz against a ~ 75 GHz range, so the conditioning supplies the entire content of the number.

Writing p_{epoch} for the chance a transmitter is on in any one epoch, a system observed in n_i independent execution blocks is sampled n_i times, so $D_i = 1 - (1 - p_{\text{epoch}})^{n_i}$; 18 of the 81 systems have $n_i > 1$. With at most three epochs at year-scale separations this is a phase-averaged bound, and an intermittent transmitter is at least as plausible *a priori*, so the $p_{\text{epoch}} = 1$ entries should never be quoted without it.

REFERENCES

- Backus P. R., Project Phoenix Team, 2002, in *Bull. Am. Astron. Soc.*, p. 1173
- Cataldi G., et al., 2023, *ApJ*, 951, 111
- Choza C., et al., 2024, *AJ*, 168, 254
- Cocconi G., Morrison P., 1959, *Nature*, 184, 844
- Czech D. J., et al., 2021, *PASP*, 133, 064502
- Drake F. D., 1960, *Physics Today*, 13, 40
- Drake F. D., 1974, in Sagan C., ed., *Communication with Extraterrestrial Intelligence (CETI)*. MIT Press
- Enriquez J. E., et al., 2017, *ApJ*, 849, 104
- Fouqué P., et al., 2018, *MNRAS*, 475, 1960
- Gaia Collaboration, Vallenari A., Brown A. G. A., et al., 2022, *arXiv:2206.05595* (*Gaia* DR3 summary)
- Garrett M. A., et al., 2026, *MNRAS*, accepted (arXiv:2605.10212)
- Gentile Fusillo N. P., et al., 2021, *MNRAS*, 508, 3877
- Hallinan G., et al., 2015, *Nature*, 523, 568
- Huang B.-L., Tao Z.-Z., Zhang T.-J., 2026, *ApJ*, 1006, 9
- Jacobson-Bell K., et al., 2025, *AJ*, 169, 233
- Johnson O. A., et al., 2023, *AJ*, 166, 193
- Li J.-K., Tao Z.-Z., Zhang T.-J., 2026, *arXiv:2603.19023*
- Ma P. X., et al., 2023, *Nat. Astron.*, 7, 492
- MacGregor M. A., Osten R. A., Hughes A. M., 2020, *ApJ*, 891, 80
- Margot J.-L., et al., 2023, *AJ*, 166, 206
- Mason L. A., Garrett M. A., Wandia K., Siemion A. P. V., 2024, *MNRAS*, 536, 2127
- Matrà L., Dent W. R. F., Wyatt M. C., et al., 2017, *MNRAS*, 464, 1415
- Dent W. R. F., et al., 2014, *Science*, 343, 1490
- Moór A., Pawellek N., Ábrahám P., et al., 2020, *AJ*, 159, 288
- Müller H. S. P., Schlöder F., Stutzki J., Winnewisser G., 2005, *J. Mol. Struct.*, 742, 215
- Morrison I. S., et al., 2023, *PASA*, 40, e038
- Poznanski D., et al., 2025, *AJ*, in press (arXiv:2505.03927)
- Price D. C., et al., 2020, *AJ*, 159, 86
- Sheikh S. Z., 2020, *Int. J. Astrobiol.*, 19, 237
- Sheikh S. Z., et al., 2025, *AJ*, 169, 108
- Tarter J., 2001, *ARA&A*, 39, 511
- Tremblay C. D., et al., 2024, *PASA*, 41, e004
- Valente G., et al., 2025, *Acta Astronautica*, in press
- Vidal C., et al., 2026, *arXiv:2605.21093*
- Pickett H. M., et al., 1998, *J. Quant. Spectrosc. Radiat. Transfer*, 60, 883
- White G. J., 2026, *MNRAS*, submitted
- Wright J. T., Kanodia S., Lubar E. G., 2018, *AJ*, 156, 260
- Zhang Z.-S., et al., 2020, *ApJ*, 891, 174